

Reading the Shape of News

Measuring narrative geometry to predict future illiquidity

Andreas Koulopoulos (2897428)

Abstract

Investors read the same public news differently, and that disagreement is hard to measure before it surfaces in trading. This thesis proposes an ex-ante measure of it, derived from the geometry of a firm's news in embedding space rather than from realized trades, and asks whether it can forecast firm-level illiquidity. Across US equities over 2016–2026, shifts in the geometry of a firm's news cloud predict residual illiquidity 14–21 weeks ahead, beyond market controls. The effect is selective—concentrated in a few firms and dated narrative events—but it specifically loads on the adverse-selection component of the spread: it appears in the quoted spread and in spread-per-volume (SV), a novel volume-normalized target, while the volume-scaled Amihud price-impact proxy remains null. Narrative geometry thus carries an early, interpretable trace of adverse-selection pressure.

Contents

1	Introduction	4
2	Literature Review	5
2.1	Asymmetric information and microstructure	5
2.2	Disagreement and Belief Modeling	5
2.2.1	Disagreement and the differential interpretation of public signals	5
2.2.2	Belief modelling and the foundations of disagreement	6
2.2.3	Empirical disagreement measures	7
2.3	Textual finance and narrative measurement	8
2.3.1	Scalar-collapse era	8
2.3.2	Topic-distribution era	8
2.3.3	Embedding era	9
2.4	Geometry and narrative structure in high-dimensional data	9
2.5	Illiquidity Measures	10
2.6	Where the thesis fits	10
3	Data	11
3.1	News data	11
3.2	Market data	11
4	Measuring Narrative Geometry	12
4.1	From text to embeddings	13
4.2	From embeddings to a narrative manifold	14
4.2.1	Temporal frame construction	14
4.2.2	Financial-discourse projection	15
4.2.3	Three-layer hierarchical narrative clustering	15
4.3	From the manifold to geometric features	17
4.3.1	Per-frame narrative state estimation	17
4.3.2	Geometric feature computation	19
5	Empirical Design	20
5.1	Liquidity target construction and market controls	20
5.2	Design matrix assembly	23
5.3	Predictive feature set and model specifications	24
5.4	Statistical operations	25
6	Results	26
6.1	Predictive content of the narrative representation	27
6.1.1	Where the signal appears	27
6.1.2	Where the signal lives	27
6.2	Horizon structure and liquidity channel	29

6.2.1	Lead time	29
6.2.2	Which liquidity channel	29
6.3	Narrative episodes	30
6.3.1	Predictive episodes	30
6.3.2	The narrative geometry behind the episodes	31
7	Discussion	31
A	Memory-Window Calibration	33
B	Financial-Discourse Projection: Details	34
C	Three-Layer Hierarchical Clustering: Details	35
D	Headline Preprocess Prompt	37

1 Introduction

According to Denzin [1989], a narrative is a story of a sequence of events that carries significance for both the narrator and the audience. We adopt this definition deliberately since it locates meaning not in static points but in the trajectories of events and the relations among them. A narrative is, therefore, a structured object, and its structure (how concentrated it is, how it is organized, and how quickly it rearranges itself) is itself information.

At the same time, these stories are not pure reflections of an underlying economic reality; they actively shape it. Shiller [2019] argues that popular narratives convey financially meaningful information and drive major economic and financial events. In order to abstract and understand this self-iterating mechanism, we consider narratives – and, more generally, public information as a higher-order space in which narratives exist – the *recording layer* of all dynamic economic processes. Inspired by Deleuze and Guattari [1983], for whom a recording/registration surface appropriates all of its generative processes (that lead to the projection onto that surface) and also the organs/means of production (which enable the realization of the process), “acts almost as the cause and the results, diverting the movement back to the processes and the process organs from which it arose.” Drawn back to our financial register, the public narrative surface inscribes the underlying productive and informational dynamics of a firm and its market. Navigating this projected reality, agents adjust their expectations and act on what is inscribed rather than on the process directly; finally, the surface loops back and conditions the dynamics it records. The narrative layer is, from this perspective, neither epiphenomenal nor merely descriptive; it is the registration surface on which economic life becomes legible and through which it is partly steered.

Before acting, agents must navigate this surface and turn it into something they can trade on. That navigation does not always yield a stable reading. When the record is concentrated and coherent, agents reach a consensus; when it is fragmented or spread across many directions at once, their interpretations diverge. The surface, in other words, can be a site of sense-making or a site of disagreement, and the shape it takes is not incidental to the news but a measurable property of it. This is the opening the thesis pursues: that the disagreement provoked by a firm’s news leaves a trace in the *geometry* of that news, and that this trace can be quantified. We ask whether a more fragmented landscape today shows up as a worse market for the firm tomorrow, and we propose a way to measure it. Rather than compress news into a sentiment score or a fixed set of topics—representations that discard exactly the structure we care about—we treat a firm’s news as a cloud of meaning that drifts and reshapes through time, and we read how hard the firm is to interpret from the shape of that cloud: how tight it is, how aligned, how stable.

Turning that idea into a measurement is itself the work of the thesis, and the argument is built as a pipeline: raw news becomes a moving cloud of embedded meaning, the cloud becomes a sequence of geometric states, those states become features, and the features are finally tested against future illiquidity. The chapters follow that same order, and each opens with a short summary paragraph to facilitate smooth “navigation”.

2 Literature Review

The thesis sits at the intersection of four literatures that have so far evolved almost independently: the microstructural theory of illiquidity that arises from disagreement about public information; the disagreement literature and belief modeling; textual finance, which quantifies free-text data to pull economic signals out of news; and the econophysics and machine-learning work that measures the geometric structure of financial data. Each supplies one building block of the argument. The contribution here is to tie them together: using the geometry of news-embedding clouds, we build an *ex-ante* proxy for the interpretability of public information and test it against the subsequent illiquidity of firms’ stocks in a 10 year panel of US equities.

2.1 Asymmetric information and microstructure

The theory of asymmetric information is the foundation for how information frictions produce observable illiquidity. In the canonical sequential-trade model of Glosten and Milgrom [1985], a risk-neutral market maker who cannot tell informed counterparties from uninformed ones posts a positive bid–ask spread purely to cover adverse selection; Kyle [1985] provides the complementary price-impact formulation, in which prices move in proportion to order flow. Later work turns the latent probability of informed trading (PIN) into an empirical measure [Easley et al., 1996, 2002] and, at high frequency, into the VPIN measure of order-flow toxicity [Easley et al., 2012]. In all of these models the “informed” counterparty holds better *private* information, so public disclosure should resolve frictions, not amplify them.

A second stream documents an empirical paradox: public news often *widens* spreads rather than narrowing them. Kim and Verrecchia [1994] traces the mechanism. When investors differ in how well they can interpret a public earnings announcement, the announcement produces “informed judgments” among the more sophisticated agents, raises bid–ask spreads, and lifts trading volume at the same time; lower liquidity and higher volume coexist around earnings releases as a direct consequence of the disclosure, not of separately arriving private information. Kim and Verrecchia [1997] sharpens the point three years later by separating private information held before the announcement from informed judgments formed during the event. Crego [2020] adds a different channel: a public signal lowers uncertainty and so draws risk-averse informed traders in more aggressively, making the order flow more “toxic”. He finds exactly this in oil-related equities, where the bid–ask spread roughly doubles and volume rises by 32% around scheduled weekly oil-inventory releases — a pattern unrelated to the surprise content of the report itself. Both papers point the same way: public news is a state in which adverse selection *intensifies* rather than dissipates.

2.2 Disagreement and Belief Modeling

2.2.1 Disagreement and the differential interpretation of public signals

A parallel theoretical literature argues that the friction following public information need not be adverse selection in the private-information sense at all; it can instead be *disagreement* about

how shared information should be read. Kandel and Pearson [1995] shows this both in theory and in the data: using I/B/E/S analyst earnings forecasts, they find that traders who see the same announcement revise their valuations in ways no common information-processing rule can reconcile, and that those divergent revisions show up as trading volume. Harris and Raviv [1993] build a model in which common information and common priors still generate volume, because agents disagree about the mapping from signals to states, and Cao and Ou-Yang [2009] carry the channel into equity and option volumes. Banerjee and Kremer [2010] show in a dynamic setting that interpretive disagreement generates the volume autocorrelation and the volume–volatility correlation we observe in the data, and Hong and Stein [2007] pull the whole literature together into a “disagreement and the stock market” view in which heterogeneous interpretation, not private information, is the primary driver of trading and prices.

2.2.2 Belief modelling and the foundations of disagreement

Beneath disagreement lies a prior question: how are beliefs formed? The classical paradigm assumes a common prior, and agents who share that prior and update by Bayes’ rule on identical information cannot “agree to disagree” [Morris, 1995]. On that assumption public information pushes toward consensus, and disagreement is only a temporary by-product of incomplete information diffusion. This thesis starts from the opposite premise, following a strand of belief modelling that relaxes exactly this assumption.

The sharpest version of this idea is Acemoglu et al. [2006]. In their Bayesian-learning model two agents observe the same infinite sequence of signals about an underlying parameter, but their uncertainty is not about the signal realizations; it is about the *mapping* from signals to states — the conditional distribution of signals given the parameter. They prove that the agents never reach consensus on the parameter, even asymptotically, and that new data can fail to pull their posteriors together and may even drive them apart.

The benchmark it overturns is the classical Bayesian one of *asymptotic agreement*, where two agents who differ only in their priors watch their posteriors converge as the shared record of observations grows, so that enough data eventually washes the disagreement out. Acemoglu et al. [2006] show how fragile this benchmark is. Once agents are merely *almost* sure of the signal-to-state mapping rather than perfectly sure — the case they call “approximate certainty” — asymptotic agreement survives, even in the limit as the residual uncertainty vanishes, only when the signal distributions have rapidly-varying tails (normal or exponential, say); it *fails* for regularly-varying tails (Pareto, log-normal, Student-*t*). On this view disagreement is not a residue that disclosure erases but a stable consequence of heterogeneous interpretation, one that more information can sustain or even widen. This is the micro-foundation for what the thesis sets out to measure: *interpretive difficulty*, the property of a public-information environment that makes the signal-to-state mapping ambiguous and belief convergence fragile.

Two further modelling approaches tie belief heterogeneity to pricing and trading. Banerjee [2011] builds a dynamic framework that nests rational expectations (RE) and difference-of-opinion (DO) learning and shows, using analyst dispersion and volume as disagreement prox-

ies, that when investors condition on prices (RE), disagreement relates positively to expected returns, return volatility, and market beta but negatively to return autocorrelation; when they do not use prices (DO), the relations flip. Atmaz and Basak [2018] push this into a tractable continuous-time equilibrium with a continuum of investors holding different beliefs, where belief dispersion raises stock volatility and trading volume in every state, prices are convex in cash-flow news, and the sign of the dispersion–return relation depends on whether the optimism or the uncertainty effect dominates. Epstein and Schneider [2008] work at a more conceptual level: ambiguity-averse agents who cannot judge the *quality* of a public signal evaluate it under a worst-case interpretation, react asymmetrically to news, and demand a premium on assets whose information environment is poor. None of these frameworks, though, offers a way to quantify how hard a given information set is to interpret without leaning on the trading behaviour it later produces. That is the gap the geometric formulation is meant to fill.

2.2.3 Empirical disagreement measures

The empirical literature has settled on a handful of disagreement proxies, all read off market outcomes after the fact. The most common is analyst forecast dispersion. Diether et al. [2002] find that high-dispersion stocks underperform low-dispersion ones by about 9.5% a year, an effect concentrated in small, recently poorly-performing stocks and consistent with the short-sale-constrained price-optimism logic of Miller-type models. Carlin et al. [2014] build a more direct measure — disagreement among Wall Street mortgage-backed-security dealers about prepayment speeds — and find that higher disagreement goes with higher expected returns, higher return volatility, and larger volume, with the evidence favouring a rational-expectations over a difference-of-opinion mechanism in that market. Huang et al. [2024] add that the *dimension* of disagreement matters: when investors disagree about the precision of public signals rather than their content, pre-announcement price efficiency falls while post-announcement price discovery and volume rise. Closest to this thesis, Li and Luan [2025] construct a firm-level, news-based disagreement measure as the elasticity of trading volume to return volatility around firm news events, extending Bollerslev et al. [2018], who use the same volume–volatility elasticity around macroeconomic announcements as a market-level proxy for differences of opinion. Li and Luan [2025] report a negative cross-sectional relation between their measure and future returns, in line with the Atmaz and Basak [2018] prediction for the regime where optimism dominates uncertainty.

These measures share one limitation: they are all *ex-post*. Analyst dispersion summarises beliefs after the forecasts are published; volume–volatility elasticity is computed from trades that have already happened; dealer surveys are run after positions are taken. None of them observes the property of the information environment that generates the disagreement — only its consequences, once the market has responded. The mechanism is well understood and microfounded; the measurement is purely retrospective. The thesis aims at this gap, computing a proxy for interpretive difficulty straight from the geometry of the news-embedding cloud, before any trading response has occurred.

2.3 Textual finance and narrative measurement

A large body of work establishes that financial text carries economically meaningful information. Its methodological reflex, though, has been to *compress* that text — into a scalar tone, a single uncertainty index, or a low-dimensional topic distribution — before running any inference. The thesis argues that this compression throws away exactly the structure needed to see interpretive difficulty, and the review below is organized to make that case.

2.3.1 Scalar-collapse era

The dictionary tradition reduces text to a single number. Tetlock [2007] measures media tone as the fraction of negative words in the Wall Street Journal’s “Abreast of the Market” column over 1984–1999 and shows, in a VAR, that high media pessimism predicts downward pressure on prices followed by reversion to fundamentals, and that unusually high or low pessimism forecasts heavier trading volume. Loughran and McDonald [2011] show that domain-general word lists misclassify a large share of the words they flag as negative in financial disclosures, and build the finance-specific lexicon that later work has adopted as the standard scalar-tone tool [Loughran and McDonald, 2016]. The same logic scales up to the macro level: Manela and Moreira [2017] construct NVIX, a text-based implied-volatility index from WSJ front-page articles reaching back to 1890, and Baker et al. [2016] build the EPU index from newspaper counts of economy-policy-uncertainty co-occurrences. Each of these captures economically interesting variation — and each collapses the full information content of news into one time series before any inference is run.

2.3.2 Topic-distribution era

A more recent literature keeps a low-dimensional distribution instead of a scalar. Shiller [2017, 2019] supply the conceptual frame, arguing that popular narratives spread through populations like epidemics and drive macroeconomic and financial events. Bybee et al. [2024] put this to work at scale, estimating a topic model on 800,000 Wall Street Journal articles from 1984–2017, extracting an interpretable set of news-attention themes, and showing that the resulting attention shares track a wide range of economic activity and add real explanatory power to a vector autoregression of macroeconomic dynamics. Bybee et al. [2023] take the method into asset pricing, building narrative pricing factors from topic loadings whose out-of-sample Sharpe ratios rival characteristic-based factors. Ke et al. [2019] go supervised instead: predictive screening picks sentiment-relevant words, a topic model assigns return-tuned weights, and penalised aggregation yields an article-level sentiment score that beats commercial sentiment products on the Dow Jones Newswires. This is the frontier of the scalar/low-dimensional tradition — and it still collapses the embedding cloud into a topic loading or a return-tuned scalar before inference.

2.3.3 Embedding era

An adjacent literature shows that pre-trained language models produce dense embeddings whose *geometry* carries information the scalar and topic methods throw away. Ethayarajh [2019] documents that contextual embeddings from BERT, ELMo, and GPT-2 are strongly anisotropic, sitting in narrow cones rather than filling the sphere, and Mu and Viswanath [2018] show that subtracting the embedding mean and top principal directions — the “all-but-the-top” correction — materially improves downstream performance. In finance, contextual embeddings are spreading fast. Domain-adapted FinBERT [Araci, 2019, Liu et al., 2020, Yang et al., 2020] is now a standard input feature; Chen et al. [2025] find that LLM embeddings of news add return-prediction power beyond traditional signals; and Gabaix et al. [2025] bring embedding-based representations of assets into mainstream asset pricing, showing that contextualized embeddings predict institutional portfolio decisions better than the alternatives, while investor embeddings open new ways to detect “crowded trades” and evaluate manager performance. Closest in spirit, Miori and Petrov [2024] show that dynamic narrative networks built from WSJ text undergo fragmentation episodes that line up with major market dislocations — direct evidence that narrative structure in news-embedding space matters financially. Yet even this most recent work consumes the embedding cloud as a feature vector or as input to a clustering step. The *geometry* of the cloud itself — its concentration, dispersion, drift, and curvature — has not been treated as a time-varying financial object and measured directly. The thesis closes that gap.

2.4 Geometry and narrative structure in high-dimensional data

A third literature, from econophysics and applied geometry, shows that the geometric structure of high-dimensional financial data has measurable economic content — but it has so far been studied on price and return data, never on text. Two strands matter here. The first is the econophysics work on correlation structure, which established that a set of spectral features — the bulk eigenvalue distribution against random-matrix benchmarks, eigenvector localization, the participation ratio, and the weight of the leading eigenvalues — can separate informative from noisy structure in financial covariance matrices [Laloux et al., 1999, Plerou et al., 1999]. These features ground the per-narrative spectral-anisotropy measure (the λ_2/λ_1 eigenvalue ratio) that the thesis applies to its narrative subclouds [Roy and Vetterli, 2007]. The second strand, subspace geometry, treats the dominant directions of a covariance matrix as a point on a Grassmann manifold, so that the principal angles between subspaces estimated in adjacent periods measure how fast the leading signal directions rotate [Edelman et al., 1998]; this is the basis for the thesis’s per-narrative directional-drift feature. Both rest on standard foundations: directional statistics on the unit hypersphere [Banerjee et al., 2005, Sra, 2012] and the Grassmann geometry of principal angles [Edelman et al., 1998]. What stands out is that this toolkit has been applied almost only to return-space objects — correlation graphs of stocks, point clouds of index returns, networks of price co-movement. No prior work projects a corpus of firm news onto the unit hypersphere and tracks, over time, the per-narrative directional and

spectral geometry of the resulting embedding manifold. The thesis redirects the toolkit from return space into news-embedding space.

2.5 Illiquidity Measures

The dependent variable against which the geometric proxy is tested is firm-level illiquidity, and the choice of proxy is not incidental. The mechanism of Section 2.1 is one of adverse selection: a liquidity supplier who cannot read the informational content of order flow protects herself by *widening the bid–ask spread* [Glosten and Milgrom, 1985]. The variable that tests that mechanism most directly is therefore the spread itself, not a generic illiquidity index.

The theory-anchoring measure is the relative quoted spread, the literal decision variable of the Glosten–Milgrom market maker, and the canonical priced measure of the liquidity premium [Amihud and Mendelson, 1986]. Microstructure theory decomposes the spread into an adverse-selection component, plus inventory-holding and order-processing components [Glosten and Harris, 1988, Huang and Stoll, 1997]; the interpretive-difficulty hypothesis is a statement about the *adverse-selection* component specifically, as it is the part of the spread that responds to the market maker’s inability to read order flow. A complication, however, is that the disagreement literature predicts that interpretive difficulty raises spreads *and* trading volume simultaneously [Kim and Verrecchia, 1994, Kandel and Pearson, 1995]: a wide spread accompanied by a volume surge is the ordinary signature of high-volume disagreement trading, not necessarily of the adverse-selection-driven widening that the hypothesis targets. This motivates the thesis’s novel construct and primary empirical target, *spread-per-volume* (SV), which normalizes the quoted spread by (log) dollar volume. By netting out the volume dimension, SV isolates the configuration that the hypothesis actually predicts – a spread the market maker charges as compensation for adverse-selection risk that is *not* underwritten by commensurate two-sided trading – and so emphasizes the adverse-selection component while de-weighting the order-processing and benign-disagreement-churn components that scale with volume. In this respect, SV inverts the logic of the existing disagreement measures: where proxies such as the volume–volatility elasticity of Li and Luan [2025] read disagreement *from* trading volume, SV divides volume out, retaining only the adverse-selection widening that volume-based measures cannot separate from benign churn. The *third* proxy, the Amihud price-impact ratio [Amihud, 2002] is retained as a robustness check. It captures Kyle’s [1985] price-impact (λ) view of illiquidity rather than the Glosten–Milgrom spread view, so the consistency of results across the spread, SV, and Amihud proxies guards against any single measure’s idiosyncrasies.

2.6 Where the thesis fits

Together, the four literatures offer complementary yet disjointed building blocks. Microstructure theory provides the mechanism: a market maker with an inability to interpret the information content of order flows increases the bid–ask spread to protect against adverse selection, and hence the information friction manifests itself in the form of illiquidity. The disagreements and belief-modeling literature identifies the *source* of such friction in misinterpretation — the

differences among investors in their interpretation of a common public signal — and attributes it to uncertainty in the interpretation of the signal-to-state map [Acemoglu et al., 2006], but measures the disagreement ex post through market realizations. Textual finance literature has found evidence for the presence of economic information in news — but condenses news information into a single scalar tonal score, a static topic loading, or even a return-tuned feature, without interpreting the signal at all. And the geometric approaches to finance have demonstrated that there is information in the spectral and directional structure of high-dimensional return data — but have applied this toolkit mostly to prices and networks.

The thesis builds an ex-ante proxy for the interpretive difficulty of public information — grounded in the Acemoglu et al. [2006] result that interpretive uncertainty is what binds belief convergence — and tests whether that proxy predicts future firm-level residual illiquidity across a panel of U.S. equities over 10 years, after controlling for VIX, realized volatility, and aggregate market liquidity. This thesis has three close counterparts. Li and Luan [2025] builds a firm-level, news-based measure of disagreement ex post using volume-volatility elasticity, while here we provide an ex-ante measure from the geometry of the news embeddings. Bybee et al. [2023] infers pricing factors from the news topic loadings, but here we retain the within-topic news embedding geometry that they discard. Also, the econophysics literature infers economics from the spectral and subspace geometry of return covariance matrices [Laloux et al., 1999, Roy and Vetterli, 2007], while here we infer from news-embedding manifolds. What’s new about this work is its synthesis of these efforts: an ex-ante, geometric, narrative-level measure of a property of information known to be important but not previously observed.

3 Data

3.1 News data

The study uses a firm-level news corpus covering the 2016–2026 period, spanning a panel of large U.S. equities. The data were gathered from the [Financial Modeling Prep] (FMP) API and the FNSPID Hugging Face dataset (Dong et al. [2024]). After fetching, we proceed with a cleaning step using an LLM call (model: deepseek-v4-flash) (Appendix D), which removes low-information material such as generic price commentary, repeated market chatter or click-bait. What remains is a set of headline statements intended to preserve operational, strategic, regulatory, and causal information.

3.2 Market data

All firm-level and market-wide price and volume data are sourced from the CRSP daily stock file via WRDS. Three illiquidity proxies are constructed from these data (full construction details in Section 5.1). The **relative quoted spread** (primary proxy) is computed from daily

best bid B_d and ask A_d quotes as

$$QS_d = \frac{A_d - B_d}{M_d}, \quad M_d = \frac{1}{2}(A_d + B_d), \quad (1)$$

retained on days where $A_d, B_d, M_d > 0$. The **spread-per-volume** (SV), the thesis’s novel construct and primary empirical target, normalises the quoted spread by log dollar volume,

$$SV_d = \frac{QS_d}{\log(1 + P_d V_d)}, \quad (2)$$

where P_d and V_d are the CRSP adjusted closing price and share volume. The **Amihud illiquidity ratio** [Amihud, 2002] (robustness proxy) is

$$ILLIQ_d = \frac{|R_d|}{P_d V_d}, \quad R_d = \log(P_d / P_{d-1}), \quad (3)$$

computed from CRSP adjusted OHLCV data; zero-volume days are set to **NaN** and the series is winsorised at its 99.5th percentile.

Three market-state controls are constructed from the same source alongside aggregate data. The **CBOE VIX** daily close is obtained from CRSP (via WRDS) and captures forward-looking aggregate uncertainty [Whaley, 2000]. A **21-day trailing realised volatility** for each firm is

$$RV_d^{(i)} = \sqrt{\frac{1}{21} \sum_{j=0}^{20} R_{d-j}^2}, \quad (4)$$

constructed from CRSP daily log-returns in the spirit of Andersen et al. [2003]. A **market-wide liquidity factor** is proxied by the Amihud ratio of SPY (Eq. 3 applied to the SPY ETF daily series from CRSP), following the aggregate liquidity tradition of Pástor and Stambaugh [2003]. All daily series are aggregated to the weekly frame grid by robust median; the three controls are then used to orthogonalize each illiquidity proxy, isolating the idiosyncratic firm-level component on which the disagreement mechanism is tested.

4 Measuring Narrative Geometry

This section builds the thesis’s central instrument: a per-firm, per-narrative measurement of the *geometry* of a firm’s news in embedding space, computed entirely ex ante from text. The construction runs as a pipeline in three stages — from raw text to semantic embeddings, from embeddings to a time-indexed narrative manifold, and from that manifold to the geometric feature series. The empirical design that tests these features against future illiquidity is deferred to Section 5.

4.1 From text to embeddings

The current block transforms a raw firm-level news corpus into a clean collection of semantic observations: headlines undergo a process of extracting only factual firm-level information, are encoded as dense vectors, and undergo the deduplication step so that multiple coverage of the same underlying information will not interfere with the geometry. The raw corpus consists of firm-level news articles for the panel tickers, collected between January 2016 and May 2026. Each article is represented by its headline and a short accompanying snippet. Cleaning is performed by a large language model (DeepSeek-V4-Flash) operating on each headline–snippet pair independently.

Filtering rule Each observation is classified as *kept* or *discarded*. An item is kept only if it satisfies all of the following:

- it contains at least one concrete business fact attributable to the main entity (operational metrics, strategic actions, regulatory events, executive changes, supply-chain dynamics, or product milestones);
- it preserves causal or temporal structure that anchors the fact (e.g. *tariff increase* $\rightarrow$ *cost pass-through*);
- it does not consist primarily of price commentary, analyst ratings, market-cap milestones, or generic investment advice.

Items that fail any criterion are discarded. This rule is intentionally conservative: noise that would collapse distinct semantic clusters in the embedding space is removed even at the cost of losing some borderline observations.

Semantic Embeddings Each retained headline h_i is mapped to a dense vector $\mathbf{e}_i \in \mathbb{R}^{3072}$ using the Google Gemini embedding model (`gemini-embedding-001`). Gemini was adopted after comparison with general-purpose and finance-adapted sentence-transformer encoders: qualitative inspection of the resulting embedding geometry showed markedly tighter intra-topic cosine clustering.

Deduplication After embeddings are computed, a greedy deduplication pass is applied within each calendar day. Headlines are sorted chronologically; each article is discarded if its cosine similarity to any already-retained headline exceeds a threshold $\tau = 0.87$. The threshold was selected through qualitative testing: a grid of candidate values $\{0.80, 0.83, 0.85, 0.87, 0.90, 0.92\}$ was evaluated by manually inspecting the retained headline sets at each level, and $\tau = 0.87$ was identified as the point at which genuine duplicates (wire-service rewrites, ticker-alert re-runs) are consistently removed while semantically distinct articles covering the same topic from different angles are preserved.

4.2 From embeddings to a narrative manifold

The block maps the embeddings to a dynamic, multi-resolution representation of the firm’s discourse. The headlines are grouped into time-decayed frames, stripped of the dominant “financial-discourse” direction that otherwise swamps thematic structure, and partitioned into three nested layers of *soft* narrative memberships—the manifold on which all geometry is subsequently measured.

4.2.1 Temporal frame construction

The second stage converts the set of headline embeddings $\mathcal{H}$ into a sequence of T temporally-indexed frames, each representing the weighted information environment on a given market date. The procedure has two steps: session mapping and exponential-decay window construction.

Session mapping Each headline is assigned to the first NYSE trading session on or after its publication date. Let $\mathcal{S} = \{s_1, s_2, \dots\}$ be the ordered set of valid NYSE sessions in the sample period. For headline i published on calendar date d_i , its session date is

$$s(i) = \min\{s \in \mathcal{S} : s \geq d_i\}. \quad (5)$$

Headlines published on non-trading days (weekends, public holidays) roll forward to the next open session. Days with no headlines are retained in the session index but contribute zero observations to any window that includes them.

Frame Definition Frames are built *per clustering layer*: each layer is scored on the frames whose memory matches its own narrative persistence, so the decay rate is layer-specific. Writing h_ℓ for the half-life of layer ℓ and $\lambda_\ell = \ln 2/h_\ell$ for the corresponding decay rate, the narrative frame $\mathcal{F}_t^{(\ell)}$ at the NYSE anchor date t collects all headlines assigned to the trading sessions in the look-back window $[t - \Delta_\ell, t]$, with a look-back of $\Delta_\ell = \lceil 4 h_\ell \rceil$ sessions — four half-lives, which hold the truncated tail mass below $1/16$. Each headline i falling in session s then receives the scalar weight

$$w_i^{(\ell)}(t) = \frac{\exp(-\lambda_\ell(t - s))}{\sum_{j \in \mathcal{F}_t^{(\ell)}} \exp(-\lambda_\ell(t - s_j))}, \quad (6)$$

with lags measured in trading sessions. The production half-lives, selected by the calibration of Appendix A, rise with layer coarseness: $h_{L1} = 35$, $h_{L2} = 53$, and $h_{L3} = 77$ trading sessions for the event, narrative, and macro-thematic layers respectively (decay rates $\lambda_\ell \approx 0.020, 0.013, 0.009$; half-lives of roughly seven, eleven, and fifteen weeks; look-backs $\Delta_\ell \approx 140, 212$, and 308 sessions).

Design note. The half-lives per layer are scaled based on their corresponding *intrinsic* persistence: event-level features have half-lives in weeks, whereas macro-thematic regimes persist over a quarter or more. As a result, the more coarse-grained the layer is, the longer the time span in the memory will be. Values $35 \rightarrow 53 \rightarrow 77$ sessions are set using the preregistered window calibration procedure of Appendix A.

4.2.2 Financial-discourse projection

Raw transformer embeddings of financial headlines are highly anisotropic: Ethayarajh [2019] find that contextualized representations tend to cluster around a single dominant direction, so unrelated sentences tend to have a high cosine similarity even if they are thematically separated (in our case, we saw a mean cosine similarity above 0.6). This rank-1 direction applied to firm-level news encodes not thematic content but firm identity and generic financial vocabulary, making naive clustering on raw embeddings almost uninformative. Thus, we apply the *all-but-the-top* correction from Mu and Viswanath [2018]: we remove the dominant direction ($\approx 73\%$ of variance, but only firm identity) from each embedding, bringing the mean cosine similarity to 0.1 and exposing the thematic structure underlying the embeddings. The discourse-removed matrix $\mathbf{X}^\perp$ is then reduced to the smallest set of k^* principal components capturing $\approx 80\%$ of its variance and projected onto the unit hypersphere, yielding the representation on which all downstream geometry is computed:

$$\bar{\mathbf{Y}} = \text{normalise}(\mathbf{X}^\perp \mathbf{V}_k) \in \mathcal{S}^{k^*-1}, \quad (7)$$

where $\mathbf{V}_k$ holds the leading k^* right singular vectors of $\mathbf{X}^\perp$. Row normalization makes the cosine-based self-tuning affinity of the next stage well-defined and motivates spherical rather than Euclidean k -means throughout. The full construction — the discourse-direction extraction, the variance-threshold rule that fixes k^* , and the frozen-geometry mode used for out-of-sample auditing — is given in Appendix B.

4.2.3 Three-layer hierarchical narrative clustering

Financial discourse runs at several time scales at once. There are macro themes, multi-week narrative arcs, and fleeting event signals, so no single clustering resolution is sufficient. This stage partitions the $\bar{\mathbf{Y}}$ cloud into three strictly nested layers (event *subset* narrative *subset* macro) of the *bar{mathbf{b}}fY* cloud from a single shared affinity graph, and assigns each headline a *soft* membership across anchors, preserving interpretive ambiguity rather than collapsing it by hard assignment. The objects that following sections consume are the resulting per-anchor soft memberships.

Design rationale. Financial discourse operates at several temporal scales at once: macro-thematic regimes (sector rotation, systemic stress) persisting over months, intermediate narrative arcs (earnings cycles, regulatory concerns) lasting weeks, and event-level signals decaying within days; ACF-based half-life estimators confirm that coarser layers persist longer [Hamilton

et al., 2016], which is exactly why each layer is read through its own matched memory window (Section 4.2.1). The three-layer design encodes this multi-resolution structure directly:

- **L3 — macro-thematic layer:** $K \in [4, 10]$ clusters.
- **L2 — narrative layer:** $K \in [9, 23]$ clusters.
- **L1 — event layer:** $K \in [17, 56]$ clusters.

Nesting is genuine and enforced throughout: every L2 cluster is a proper subset of an L3 cluster, and every L1 cluster of an L2 cluster, so that event-level signals are always interpretable within a coherent narrative context, and that context within a macro-thematic frame.

Construction. The three partitions are obtained by *divisive spectral* splitting of a single shared affinity graph over $\bar{\mathbf{Y}}$: a self-tuning cosine affinity [Zelnik-Manor and Perona, 2004] is built once, normalised to a doubly-stochastic matrix [Sinkhorn and Knopp, 1967], and spectrally decomposed. The top layer L3 is cut at the most prominent gap in the Laplacian spectrum [von Luxburg, 2007], and each parent is then recursively refined into L2 and L1 children by spherical k -means [Dhillon and Modha, 2001], the child count at every split chosen by the local eigengap within the preset bands above. Divisive splitting is preferred to agglomerative linkage, which is prone to chaining artefacts in high dimensions. The full graph construction (near-duplicate node collapse, the self-tuning affinity, Sinkhorn normalisation, and the Laplacian spectral decomposition), the eigengap rule, and the per-layer child-count bands and floors are given in Appendix C.

Anchor bases and soft membership. Each cluster k carries an *anchor basis* $\mathbf{B}_k$ — an orthonormal set of its leading within-cluster principal directions (rank selection in Appendix C) — which serves as the cluster’s local coordinate frame for the geometric features of Section 4.3. Rather than assign each headline to a single cluster, soft membership weights are derived from how well each anchor subspace reconstructs it. The residual of $\bar{\mathbf{y}}_i$ against anchor subspace $\mathbf{B}_k$,

$$r_{i,k} = \left\| \bar{\mathbf{y}}_i - \mathbf{B}_k \mathbf{B}_k^\top \bar{\mathbf{y}}_i \right\|^2, \quad (8)$$

is converted to a weight via a power law on *relative* residuals:

$$w_{i,k} = \frac{(r_{i,k} / \min_j r_{i,j})^{-\alpha}}{\sum_j (r_{i,j} / \min_j r_{i,j})^{-\alpha}}, \quad (9)$$

where normalising by $\min_j r_{i,j}$ maps the best-matching anchor to one and amplifies contrasts for worse-matching anchors. Hard assignment — taking $\arg \min_k r_{i,k}$ — would discard exactly the ambiguous headlines whose interpretive difficulty the thesis is designed to measure; the soft weights preserve that ambiguity and propagate it into the downstream geometric features. The concentration parameter α is calibrated per layer to avoid both mode collapse and over-diffusion (Appendix C).

4.3 From the manifold to geometric features

This block reads the shape of each narrative through time. For every frame and every anchor cloud, it estimates the narrative’s directional concentration, its drift, and its spectral fragmentation, producing the per-anchor geometric time series that are the actual inputs to the predictive models.

4.3.1 Per-frame narrative state estimation

This stage maps the static decomposition produced in 4.2.3 into a time-indexed panel of narrative states. For each of the T frames and each of the K anchors at the given layer, it computes four quantities: a directional concentration parameter, a weighted mean direction, a narrative activation share, and a credible interval on that share. Together, these constitute the primary time series passed to downstream geometry and regression stages.

Combined frame-anchor weights. Each frame t is defined by a window $\mathcal{W}_t$ of headlines together with scalar decay weights $\omega_{i,t}$ from the frame construction of Section 4.2.1. The narrative soft assignments $\pi_{i,k}$ from the clustering of Section 4.2.3 are combined with the temporal weights to form a joint weight:

$$w_{i,t,k} = \omega_{i,t} \cdot \pi_{i,k}, \quad (10)$$

with effective mass $m_{t,k} = \sum_{i \in \mathcal{W}_t} w_{i,t,k}$ representing how strongly anchor k is represented in frame t . The weighted resultant vector is

$$\mathbf{R}_{t,k} = \sum_{i \in \mathcal{W}_t} w_{i,t,k} \bar{\mathbf{y}}_i \in \mathbb{R}^{201}. \quad (11)$$

$\mathbf{R}_{t,k}$ is intentionally not normalised: its magnitude $\|\mathbf{R}_{t,k}\|_2$ encodes how directionally coherent the weighted subcloud is.

Mean direction and mean resultant length. The mean direction of anchor k at frame t is

$$\hat{\boldsymbol{\mu}}_{t,k} = \frac{\mathbf{R}_{t,k}}{\|\mathbf{R}_{t,k}\|_2} \in \mathcal{S}^{200}, \quad (12)$$

and the mean resultant length is the scalar

$$\bar{r}_{t,k} = \frac{\|\mathbf{R}_{t,k}\|_2}{m_{t,k}} \in [0, 1]. \quad (13)$$

When $\bar{r}_{t,k} \approx 1$, all weighted headlines point in nearly the same direction — the narrative is focused. When $\bar{r}_{t,k} \approx 0$, the headlines are dispersed uniformly on $\mathcal{S}^{200}$ and the anchor carries no coherent directional signal in that frame.

vMF directional concentration $\hat{\kappa}_{t,k}$. The mean resultant length is converted to a concentration parameter under the von Mises-Fisher (vMF) model. For a vMF distribution on $\mathcal{S}^{p-1}$, Banerjee et al. (2005) give the maximum likelihood estimator of the concentration κ as the solution to $A_p(\kappa) = \bar{r}$, where A_p is the ratio of modified Bessel functions. In the high-dimensional regime $p \gg 1$ the standard approximation reduces to

$$\hat{\kappa}_{t,k} = \frac{\bar{r}_{t,k} (p - \bar{r}_{t,k}^2)}{1 - \bar{r}_{t,k}^2}, \quad p = 201. \quad (14)$$

The estimator is monotone increasing in $\bar{r}_{t,k}$: values near the ambient dimension $p = 201$ indicate moderate focus, while $\hat{\kappa}_{t,k} \gg p$ indicates near-degenerate concentration on a single direction. The estimate is set to NaN when $m_{t,k} < m_{\min} = 0.01$, marking the anchor as inactive in that frame.

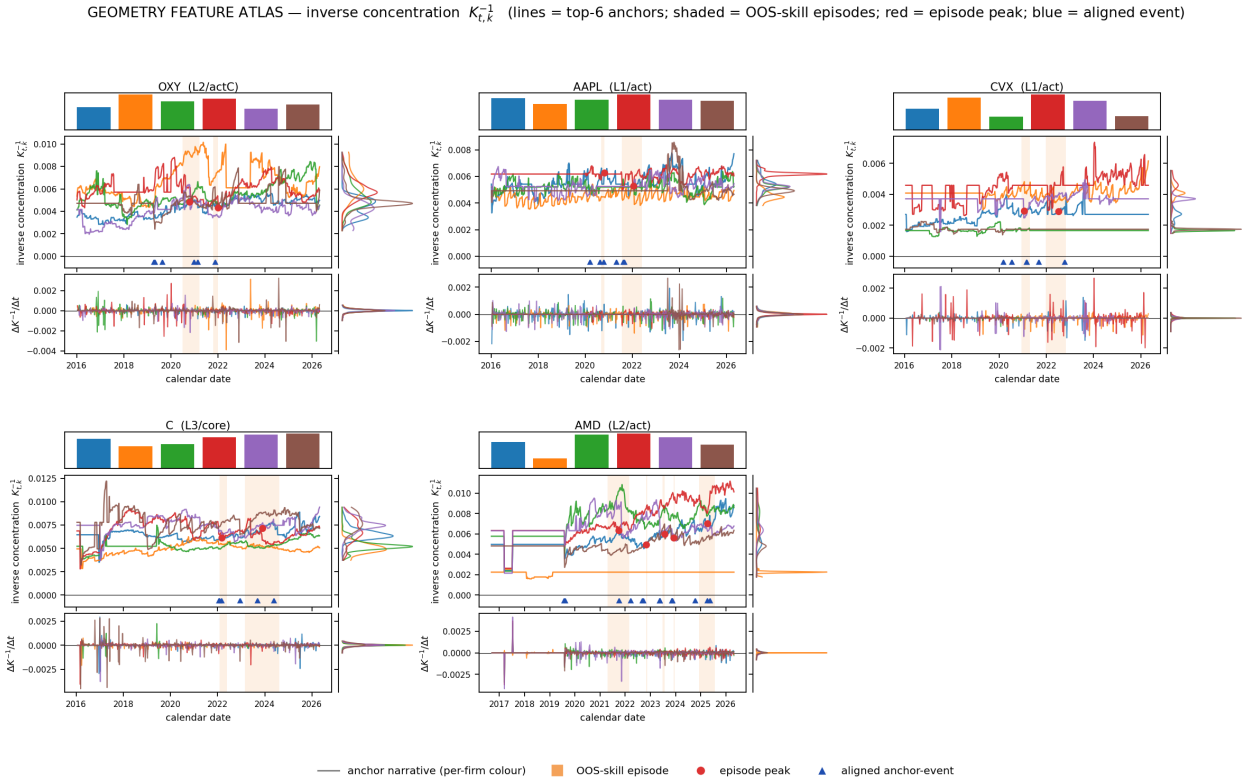


Figure 1: Per-firm feature map of the inverse directional concentration $K_{t,k}^{-1} = 1/(\hat{\kappa}_{t,k} + \epsilon)$. Each firm’s panel shows the per-anchor salience (top), the $K_{t,k}^{-1}$ trajectories with the out-of-sample predictive episodes shaded (main), the cross-sectional distribution (right), and the weekly rate of change (bottom). Large values denote a diffuse, weakly concentrated narrative; small values a focused one. The shaded *predictive episodes* are the windows in which the cell’s forecasts are accurate (Section 6.3.1), so the figure shows how movements in the geometry align with where it forecasts.

Dirichlet posterior on activation shares. Narrative activation $a_{t,k}$ is defined as the posterior mean share of anchor k in frame t under a conjugate Dirichlet model. The effective masses $m_{t,k}$ serve as weighted pseudo-counts. A symmetric Dirichlet prior with concentration

$\alpha_0 = 1.0$ (uniform) assigns prior pseudo-count α_0/K^* to each anchor. The posterior is

$$\boldsymbol{\theta}_t \mid \mathcal{W}_t \sim \text{Dir}\left(\frac{\alpha_0}{K^*} + m_{t,1}, \dots, \frac{\alpha_0}{K^*} + m_{t,K^*}\right), \quad (15)$$

with posterior mean

$$a_{t,k} = \mathbb{E}[\theta_{t,k}] = \frac{\alpha_0/K^* + m_{t,k}}{S_0}, \quad S_0 = \sum_j \left(\frac{\alpha_0}{K^*} + m_{t,j}\right), \quad (16)$$

and posterior standard deviation derived from the Beta marginal of each coordinate:

$$\text{sd}(\theta_{t,k}) = \sqrt{\frac{(\alpha_0/K^* + m_{t,k})(S_0 - \alpha_0/K^* - m_{t,k})}{S_0^2(S_0 + 1)}}. \quad (17)$$

Ninety-percent credible intervals are computed from the same Beta marginal. By construction, $\sum_k a_{t,k} = 1$ for every frame t , so the activations form a proper probability simplex. An anchor is classified as active in frame t if $a_{t,k} \geq a_{\min}$, where $a_{\min}$ is calibrated once from the 20th percentile of all non-zero activations across the full sample, yielding a value of approximately 0.03.

4.3.2 Geometric feature computation

In this phase, we map the decomposed static structures to a temporal array of per-anchor geometric signals, consisting of the spectral fragmentation ratio, the directional drift increment, and the inverse directional concentration for each of the K^* anchors in each of the T frames. These signals serve as the raw $T \times K^*$ matrix of features that give rise to the predictive feature blocks of Section 5.3.

Per-frame weighted local covariance. For each frame t and anchor k , the active subcloud is $\mathcal{C}_{t,k} = \{(\bar{\mathbf{y}}_i, w_{i,t,k}) : i \in \mathcal{W}_t, \pi_{i,k} > 0.01\}$, with combined decay–narrative weights $w_{i,t,k} = \omega_{i,t} \pi_{i,k}$ and normalised weights $\tilde{w}_{i,t,k} = w_{i,t,k} / \sum_j w_{j,t,k}$. The weighted mean and covariance of the subcloud are

$$\bar{\boldsymbol{\mu}}_{t,k} = \sum_{i \in \mathcal{C}_{t,k}} \tilde{w}_{i,t,k} \bar{\mathbf{y}}_i, \quad \boldsymbol{\Sigma}_{t,k} = \sum_{i \in \mathcal{C}_{t,k}} \tilde{w}_{i,t,k} (\bar{\mathbf{y}}_i - \bar{\boldsymbol{\mu}}_{t,k})(\bar{\mathbf{y}}_i - \bar{\boldsymbol{\mu}}_{t,k})^\top, \quad (18)$$

both computed in the reduced k^* -dimensional ambient space of $\bar{\mathbf{Y}}$ (Eq. 7). When the effective mass $m_{t,k} < m_{\min}$ or fewer than two distinct headlines populate the subcloud, the anchor is marked inactive and all three feature series receive their layer-specific imputed values (the per-anchor median over active frames) for that cell.

Spectral fragmentation $C_{t,k}^{(\lambda)}$. A truncated eigendecomposition of $\boldsymbol{\Sigma}_{t,k}$ in the anchor basis $\mathbf{B}_k$ yields the ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$. The spectral fragmentation ratio $C_{t,k}^{(\lambda)} = \lambda_2/\lambda_1$ (Eq. 31) is stored as a $T \times K^*$ array. When $\lambda_1 \leq 0$ the entry is set to zero. As a complementary

diagnostic, the effective rank

$$\text{eff_rank}_{t,k} = \exp\left(-\sum_j \tilde{\lambda}_j \ln \tilde{\lambda}_j\right), \quad \tilde{\lambda}_j = \frac{\lambda_j}{\sum_j \lambda_j}, \quad (19)$$

is recorded alongside the ratio; the joint criterion $\text{eff_rank}_{t,k} \geq 7$ and $C_{t,k}^{(\lambda)} \geq 0.45$ is used in diagnostics to flag cells exhibiting full spectral fragmentation, i.e. at least two directions carry comparable and substantial variance simultaneously. This array feeds directly into feature block $\mathbf{C}^{(\lambda)}$ of Section 5.3.

Directional drift $\Delta\phi_{t,k}$. The fixed anchor basis $\mathbf{B}_k = [\mathbf{b}_{k,1}, \dots, \mathbf{b}_{k,r_k}]$ is estimated once on the full-sample anchor subcloud (Section 4.2.3). Projecting the per-frame weighted centroid $\bar{\boldsymbol{\mu}}_{t,k}$ onto this basis gives the coordinate path $\phi_{t,k,j} = \mathbf{b}_{k,j}^\top \bar{\boldsymbol{\mu}}_{t,k}$, and its frame-to-frame increment

$$\Delta\phi_{t,k,j} = \phi_{t,k,j} - \phi_{t-1,k,j} \quad (20)$$

measures how far the narrative’s semantic centre of gravity moves along basis direction j between consecutive frames. The dominant component $j^*(k) = \arg \max_j \text{Var}_t(\Delta\phi_{t,k,j})$ is selected per anchor, and the resulting scalar series $\Delta\phi_{t,k,j^*(k)}$ is standardized by expanding-window z -score with a minimum history of 20 frames. The full $T \times K^*$ array of standardized drift increments feeds feature block $\mathbf{B}$ of Section 5.3. The use of the full-sample anchor basis as a fixed coordinate frame is a deliberate design choice: it ensures that the coordinate system is stable and interpretable across the entire sample, so that a positive $\Delta\phi$ always corresponds to movement in the same semantic direction regardless of frame.

5 Empirical Design

This section specifies how the geometric measure is tested. It constructs the liquidity targets — ordered by their alignment with the adverse-selection mechanism: the quoted spread, its volume-normalized variant SV, and the Amihud ratio — residualizes them against market controls, assembles the design matrix, and lays out the nested elastic-net specifications and the hypothesis structure through which the geometric signal is identified.

5.1 Liquidity target construction and market controls

This last pre-inference step sets the liquidity targets and the market controls to which they are orthogonalized. The pipeline creates *three* targets: the relative quoted spread (the theoretical anchor, being the literal decision variable of the Glosten–Milgrom market maker), its volume-normalized refinement SV — the thesis’s novel construct and the primary empirical target that isolates the adverse-selection component the hypothesis predicts, and finally the Amihud price-impact ratio (retained as a price-impact robustness check). All three share the same anchor grid, the same control set, the same standardization, the same orthogonalization, and the same

forecasting machinery.

Anchor grid. Targets are defined on a weekly anchor grid $\{t\}_{t=1}^T$ derived from the frame anchor dates of Section 4.2.1 by retaining dates at least seven calendar days apart. For a daily series x_d , the frame value is the *robust median* over the inter-anchor window,

$$\text{agg}_t(x) = \text{median}\{x_d : a_{t-1} < d \leq a_t\}, \quad (21)$$

where a_t is the t -th anchor date. The median (rather than a decay-weighted mean) is used throughout this stage for resilience to the heavy right tails of daily liquidity series; it is applied identically to the targets and to every control, so the dependent and control variables live on a common temporal footing.

Proxy 1 (theory anchor) — Quoted spread. From daily best bid and ask quotes (CRSP), the relative quoted spread is

$$\text{QS}_d = \frac{A_d - B_d}{M_d}, \quad M_d = \frac{1}{2}(A_d + B_d), \quad (22)$$

retained only when $A_d, B_d, M_d > 0$. The quoted spread is the proximate variable in the adverse-selection mechanism of Glosten and Milgrom [1985]: it is precisely the quantity the liquidity supplier widens when she cannot read the informational content of order flow, and the canonical direct measure of the liquidity premium [Amihud and Mendelson, 1986].

Proxy 2 (primary empirical target, novel) — Spread-per-volume (SV). The second target normalizes the quoted spread by trading activity,

$$\text{SV}_d = \frac{\text{QS}_d}{\log(1 + \text{DVOL}_d)}, \quad \text{DVOL}_d = P_d V_d, \quad (23)$$

defined when QS_d is valid and $\log(1 + \text{DVOL}_d) > 0$, and winsorized at the 99.5th percentile. SV is the pipeline’s primary novel liquidity construct, and its design is supported by the following mechanism: According to microstructure theory, the quoted spread is the sum of an adverse selection component, an inventory-holding component, and an order-processing component [Glosten and Harris, 1988, Huang and Stoll, 1997]. The interpretive difficulty hypothesis is concerned with the former. However, a simple quoted spread target is contaminated by the presence of the disagreement channel: Kim and Verrecchia [1994] and Kandel and Pearson [1995] prove that interpretive difficulty not only increases the quoted spread but also the trading volume. This means that an increase in spread accompanied by a growth in trading volume is an example of ordinary high volume disagreement trading and not the predicted adverse selection increase. The division of the spread by the log dollar volume factor controls for the latter and captures the state of interest where the spread is wide but *not* supported by two-sided trading activity— that is, a situation where the market maker charges a high per-trade premium for adverse selection risk despite low trading volume.

Proxy 3 (robustness) — Amihud illiquidity. For each trading day d of a given ticker the Amihud ratio [Amihud, 2002] is

$$\text{ILLIQ}_d = \frac{|R_d|}{\text{DVOL}_d}, \quad \text{DVOL}_d = P_d V_d, \quad R_d = \log(P_d/P_{d-1}), \quad (24)$$

computed from adjusted OHLCV data; zero-volume days are set to `NaN`, and the series is winsorized at its 99.5th percentile. The Amihud measure can be considered a robust daily frequency approximation of Kyle’s [1985] price impact [Hasbrouck, 2009, Amihud et al., 2021]; it uses the price impact (λ) concept of illiquidity instead of the Glosten–Milgrom spread, and thus the consistency of the results among the three provides a good cross-check to rule out the possibility that the geometric signal exists.

Market controls. The control vector aggregates three daily series to the anchor grid using the same median operator (21):

$$\mathbf{M}_t = [\text{VIX}_t, \text{RV}_t^{(i)}, \tilde{L}_t^{\text{SPY}}]^\top, \quad (25)$$

namely the CBOE VIX close (forward-looking market fear), the ticker’s own 21-day trailing realised volatility $\text{RV}_d^{(i)} = \sqrt{\frac{1}{21} \sum_{j=0}^{20} R_{d-j}^2}$ (backward-looking firm volatility), and the SPY Amihud illiquidity $\text{ILLIQ}_d^{\text{SPY}}$ as the common market-liquidity factor in the spirit of Pástor and Stambaugh [2003]. The SPY term enters as an expanding-window z -score $\tilde{L}_t^{\text{SPY}}$ (Eq. 26, minimum 52-frame history) so that no future information leaks into the control.

Standardisation and orthogonalised residual. Each raw target series

$$X_t \in \{\text{agg}_t(\text{ILLIQ}), \text{agg}_t(\text{QS}), \text{agg}_t(\text{SV})\}$$

is first converted to an expanding-window z -score,

$$X_t^z = z(X_t),$$

where

$$z(x_t) = \frac{x_t - \bar{x}_{1:t}}{\text{sd}(x_{1:t})} \quad (26)$$

standardizes a series using only the mean $\bar{x}_{1:t}$ and standard deviation $\text{sd}(x_{1:t})$ of frames strictly prior to t ; the score is then projected out of the span of the controls by expanding-window OLS *with intercept*. For each frame $t > t_0$ ($t_0 = 52$ frames, the one-year warm-up),

$$\hat{\boldsymbol{\delta}}_{1:t-1} = \arg \min_{\boldsymbol{\delta}} \sum_{s=1}^{t-1} (X_s^z - \delta_0 - \boldsymbol{\delta}_{1:3}^\top \mathbf{M}_s)^2, \quad X_t^\perp = X_t^z - \hat{\delta}_0 - \hat{\boldsymbol{\delta}}_{1:3}^\top \mathbf{M}_t. \quad (27)$$

The expanding window ensures $\hat{\boldsymbol{\delta}}$ uses only frames preceding t , preserving temporal precedence. The residual $X_t^\perp$ captures firm-specific liquidity conditions purged of broad market stress and

is the dependent quantity for all predictive models.

Forward targets. Two aggregation conventions are produced for each proxy: a *trailing* residual aligned to the window $(a_{t-1}, a_t]$, used as a contemporaneous state variable, and a *forward* residual aligned to $(a_t, a_{t+1}]$, used to build prediction targets. For horizons $h \in \{1, \dots, 21\}$ frames, the target is the *change* in the forward residual.

$$Y_{t,h} = X_{t+h}^{\perp, \text{fwd}} - X_t^{\perp, \text{fwd}}, \quad (28)$$

so that the models predict the *deterioration* in residual illiquidity over the next h weekly frames rather than its level. The dense horizon grid $h = 1, \dots, 21$ (roughly one to twenty-one weeks ahead) supports the horizon-profile analysis through which the predictive signal’s lead time is characterized.

5.2 Design matrix assembly

With the per-frame geometric arrays of Section 4.3.2 and the residual illiquidity targets of Section 5.1 in hand, this stage assembles the joint design matrix for the elastic-net predictive models.

Feature normalization Each raw feature series — activations $a_{t,k}$, inverse concentrations $K_{t,k}^{-1}$, drift increments $B_{t,k}$, and spectral ratios $C_{t,k}^{(\lambda)}$ — is standardized by expanding-window z -score (Eq. 26) with a minimum warm-up of 20 frames. The expanding window is essential: it ensures that neither the mean nor the standard deviation used to standardize frame t incorporates any information from frames $t + 1, \dots, T$, preserving the temporal precedence discipline throughout. Anchors that are inactive at frame t ($a_{t,k} < a_{\min}$) receive their per-anchor median as the standardized value for the concentration and drift blocks, and zero for the spectral-ratio block, so that inactivity is encoded as a neutral rather than a missing signal.

Horizon target construction and alignment For each frame t and horizon $h \in \{1, \dots, 21\}$, the forward target is the change in residual illiquidity defined in Eq. (28): $Y_{t,h} = X_{t+h}^{\perp, \text{fwd}} - X_t^{\perp, \text{fwd}}$. We align the feature matrix with the full horizon grid, producing one $(T - h_{\max}) \times P$ design matrix per layer ℓ , where P is the total number of features across all blocks for that layer. Each row of this matrix is dated by the *feature frame* t ; the corresponding target is dated $t + h$ for the chosen horizon. No row of the design matrix is constructed from a frame t for which the target $Y_{t,h}$ would require data beyond the end of the sample. The output is a set of aligned (features, targets) pairs — one pair per layer and horizon — that enter the elastic-net fitting loop of section 5.3 directly, with no further transformation.

5.3 Predictive feature set and model specifications

The locked models predict the forward target from five standardized feature blocks, each a $T \times K^*$ matrix indexed by frame t and anchor k (the market block excepted). Two of the blocks summarize the per-anchor geometry of the narrative subcloud, for which we first define the weighted local covariance. For cell (t, k) , the subcloud weights and the weighted mean and covariance $(\bar{\boldsymbol{\mu}}_{t,k}, \boldsymbol{\Sigma}_{t,k})$ are as defined in Eq. (18) of Section 4.3.2.

Block A — narrative activation. The Dirichlet posterior activation $a_{t,k}$ of Eq. (16): the salience of anchor k in frame t , irrespective of its internal geometry. This is the workhorse block, present in every specification.

Block K^{-1} — inverse directional concentration. The reciprocal of the von Mises–Fisher concentration of Eq. (14),

$$K_{t,k}^{-1} = \frac{1}{\hat{\kappa}_{t,k} + \epsilon}, \quad \epsilon = 10^{-6}, \quad (29)$$

imputed at the per-anchor median where the anchor is inactive. Large $K_{t,k}^{-1}$ marks a directionally diffuse, low-coherence narrative; the block is the natural scalar summary of within-anchor dispersion.

Block B — directional drift. Each anchor carries a fixed orthonormal basis

$$\mathbf{B}_k = [\mathbf{b}_{k,1}, \dots, \mathbf{b}_{k,r_k}] \in \mathbb{R}^{201 \times r_k},$$

estimated once on the full-sample anchor subcloud. Projecting the weighted anchor centroid $\bar{\boldsymbol{\mu}}_{t,k}$ of Eq. (18) onto this basis gives the coordinate path $\phi_{t,k,j} = \mathbf{b}_{k,j}^\top \bar{\boldsymbol{\mu}}_{t,k}$, whose frame-to-frame change

$$\Delta\phi_{t,k,j} = \phi_{t,k,j} - \phi_{t-1,k,j} \quad (30)$$

measures how fast the narrative’s semantic centre of gravity moves along basis direction j . Each $\Delta\phi_{t,k,j}$ is standardized by the expanding-window z-score of Eq. (26) (minimum twenty frames of history), and for each anchor, the single highest-variance component $j^*(k) = \arg \max_j \text{Var}_t(\Delta\phi_{t,k,j})$ is selected, so that the block collapses to one drift series per anchor $B_{t,k} = z(\Delta\phi_{t,k,j^*(k)})$. It captures the rate at which a narrative’s semantic centre of gravity moves along its dominant basis direction between consecutive frames.

Block $C^{(\lambda)}$ — spectral fragmentation. From the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the local weighted covariance $\boldsymbol{\Sigma}_{t,k}$ of Eq. (18) (computed in the anchor basis), the block records the second-to-first eigenvalue ratio

$$C_{t,k}^{(\lambda)} = \frac{\lambda_2}{\lambda_1}, \quad (31)$$

set to zero where the anchor is inactive. A value near zero indicates a unimodal, single-axis subcloud; a value approaching one indicates that a second direction carries comparable variance

— the signature of a narrative fragmenting into competing interpretations. It contributes predictive weight chiefly at the narrative (L2) resolution.

Market residualisation The three market controls $\mathbf{m}_t = (\text{VIX}_t, \text{RV}_t, L_t^{\text{SPY}})$ of Section 5.1 are *not* entered as regressors in the elastic-net models. Instead, they are projected out of the dependent variable in the expanding-window OLS step of Eq. (27) before any predictive model is fitted. The target $Y_{t,h}$ of Eq. (28) is therefore already orthogonal to broad market conditions by construction. All narrative models consequently compete against the **prevailing mean** (a zero-forecast, equivalent to the expanding-window unconditional mean of $X^\perp$) as the operative baseline.

Model specifications The blocks are assembled into a small, fixed family of nested elastic-net specifications, estimated independently at each narrative resolution (layer). Because the target $Y_{t,h}$ is already orthogonal to market conditions (Section 5.1), the natural baseline for out-of-sample R^2 is the **prevailing mean** — the expanding-window unconditional mean of $X^\perp$, equivalent to a zero forecast for $Y_{t,h}$:

$$\begin{aligned} \text{act} &: \mathbf{A}, & \text{core} &: \mathbf{A} + \mathbf{K}^{-1}, \\ \text{AB} &: \mathbf{A} + \mathbf{B}, & \text{actC} &: \mathbf{A} + \mathbf{C}^{(\lambda)}, \\ \text{full} &: \mathbf{A} + \mathbf{B} + \mathbf{K}^{-1} + \mathbf{C}^{(\lambda)}. \end{aligned}$$

A positive out-of-sample R^2 relative to the prevailing mean therefore establishes that narrative blocks carry incremental signal over and above the expanding-window mean of market-residualized illiquidity. Each specification is fitted using an expanding-window elastic net at every horizon $h = 1, \dots, 21$ of Eq. (28), with the ℓ_1/ℓ_2 mix and penalty chosen by nested cross-validation within each training origin so that no future information enters either feature standardization or hyperparameter selection. An AR(1) persistence model — regressing $Y_{t,h}$ on the lagged residual $X_t^\perp$ — is included as a secondary benchmark to establish whether any predictive gain derives merely from illiquidity autocorrelation rather than from the narrative features themselves.

5.4 Statistical operations

The claim is that the *shape* of a narrative’s embedding subcloud carries information about how investors interpret a firm-specific risk, and that this shape moves before the risk surfaces in observable illiquidity. The inference separates three kinds of content: *activation* (how salient a narrative is, regardless of its shape), *geometry* (the shape itself), and *market regime*.

Formal hypothesis structure The thesis tests one claim in two parts. First, the *geometry* of a firm’s news cloud predicts its future illiquidity (H_1). Second, it predicts through the *adverse-selection* component of illiquidity specifically (H_2). The design tests each in turn.

The null is $H_0 : \eta_k = 0 \forall k$, where η_k is any coefficient on a geometric feature of anchor k . The target is the forward residual-illiquidity change $Y_{t,h}$ of Eq. (28). Because $Y_{t,h}$ is the change in the market-orthogonalized residual $X^\perp$ (Eq. (27)), market conditions — VIX, realized volatility, aggregate liquidity — are already projected out before any model is fitted. There is no market-factor model to beat, only two benchmarks: the **prevailing mean** (a zero forecast, the expanding mean of $X^\perp$) and an **AR(1) persistence** model (regressing $Y_{t,h}$ on $X_t^\perp$).

H_1 (**prediction**) The geometric features — directional drift $\mathbf{B}$, spectral fragmentation $\mathbf{C}^{(\lambda)}$, and inverse concentration $\mathbf{K}^{-1}$ — predict $Y_{t,h}$ beyond both benchmarks and beyond mere narrative activation $\mathbf{A}$. The test is whether the *shape* of discourse, not just its prominence, carries signal. This is what separates the measure from scalar sentiment [Tetlock, 2007], topic-factor [Bybee et al., 2023], and ex-post disagreement [Diether et al., 2002, Li and Luan, 2025] constructs.

H_2 (**channel**) Given H_1 , the predictable variation sits in the *adverse-selection* component of the spread. The spread has three parts: adverse selection, inventory, and order processing [Glosten and Harris, 1988, Huang and Stoll, 1997]; disagreement raises spreads and volume together [Kim and Verrecchia, 1994, Kandel and Pearson, 1995]. We read the channel off where the signal loads across the three proxies. It should be positive in the quoted spread and in spread-per-volume (SV), which divides volume out and keeps only the adverse-selection widening, and null in the volume-scaled Amihud price-impact ratio.

The proxy comparison is therefore the test of H_2 , not a robustness check: a positive spread/SV result with a null Amihud is the signature of adverse selection, not of the high-volume churn that ordinary disagreement trading also produces. H_1 establishes that narrative geometry forecasts firm-specific illiquidity; H_2 attributes that forecast to adverse selection. This is the single message the thesis sets out to validate.

6 Results

This section asks how well the news-geometry features forecast residual stock illiquidity $X_\perp$, and where that forecasting power comes from. Unless stated otherwise, every statistic is computed on the pre-registered confirmation band $h \in [14, 21]$ weeks, with the spread-per-volume (SV) proxy as the primary illiquidity target, rolling-origin elastic-net forecasts, and the expanding in-sample mean as the benchmark. We summarise predictive accuracy by the out-of-sample (OOS) R^2 against that benchmark: a positive value means the narrative geometry beats a naive forecast that simply carries the past mean forward. Significance is judged by the stationary block bootstrap (SBB, $B = 2000$), and family-wise error is held in check by the Romano–Wolf step-down procedure; a cell tagged **ROBUST** survives that correction. The evaluation panel holds 341 firm–layer–specification cells with a valid band statistic, spanning 37 firms and the three clustering layers $L1$ (fine), $L2$ (medium), and $L3$ (coarse).

The findings follow the two hypotheses. Section 6.1 tests H_1 : how much the geometry predicts across the panel, and which firms carry the signal. Section 6.2 tests H_2 : the liquidity

channel the signal works through, together with its lead time and robustness. Section 6.3 then shows *when* the signal fires and the narrative geometry behind those windows.

6.1 Predictive content of the narrative representation

6.1.1 Where the signal appears

Across the panel there is no predictability on average. The band R^2 has a median of -0.0002 and a maximum of $+0.219$, and 166 of the 341 cells (49%) clear zero. The distribution is sharply right-skewed: most cells sit on the no-predictability benchmark, while a small group reaches materially positive accuracy. Table 1 breaks this down by clustering layer. No single layer dominates: the largest effects sit at $L2$ and $L3$, several smaller robust effects at $L1$. In all, 16 cells are ROBUST, 9 BORDERLINE, and 316 NOT_SIGNIFICANT; 25 reject the no-predictability null at SBB $p < 0.05$ and 51 at $p < 0.10$.

Table 1: Band R^2 panel summary by clustering layer (SV proxy, $h \in [14, 21]$, expanding-mean benchmark, SBB $B = 2000$). “Positive” counts cells with $R^2 > 0$; “Robust” counts cells surviving Romano–Wolf control. Medians are reported in preference to means, which are distorted by a small number of large-negative cells.

Layer	Cells	Positive	Median R^2	Max R^2	SBB $p < .05$	Robust
$L1$	93	48	+0.0015	0.1615	6	6
$L2$	155	76	-0.0002	0.2190	11	4
$L3$	93	42	-0.0013	0.1571	8	6
All	341	166	-0.0002	0.2190	25	16

6.1.2 Where the signal lives

The cells that survive inference cluster in just nine firms. Citigroup carries seven, AMD six, Apple four, and Occidental and Meta two each. Figure 2 shows the best cell per firm as a forest plot with SBB-bootstrap confidence intervals. The predictive content lives at specific horizons within the band, not in their average, so the markers are coloured by *within-band horizon* significance: blue when at least one horizon $h \in [14, 21]$ survives Romano–Wolf family-wise control ($p < 0.05$), amber when a horizon is significant under the block bootstrap but does not survive correction, grey otherwise; the count beside each label is the number of significant horizons. Fifteen of the thirty-one firms meet the family-wise criterion, against nine on the band-averaged statistic, so band averaging washes out part of the horizon-specific skill. Table 2 then lists, for every firm with at least one cell at SBB $p < 0.05$ on the band statistic, its strongest cell and the band R^2 that same cell delivers under the quoted-spread and Amihud proxies.

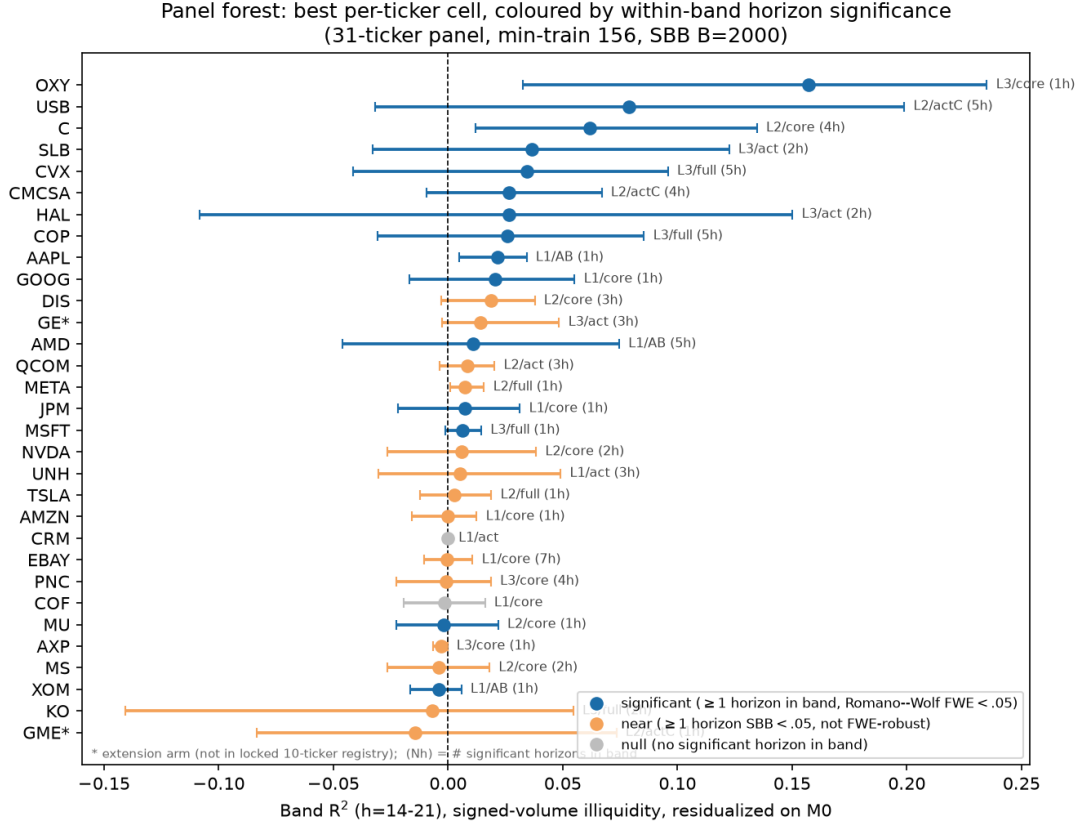


Figure 2: Panel forest of the best cell per firm (band R^2 , $h \in [14, 21]$, SV proxy, residualised on the market-control block $\mathbf{M}$). Point estimates are shown with SBB confidence intervals. Colour denotes *within-band horizon* significance rather than band-level significance: blue marks firms with at least one horizon $h \in [14, 21]$ surviving Romano–Wolf family-wise control ($p < 0.05$), amber marks firms with a nominally significant horizon (block-bootstrap $p < 0.05$) that does not survive correction, and grey marks firms with no significant horizon. The parenthetical “(Nh)” beside each label is the number of significant horizons in the band.

Table 2: Best significant cell per firm (SBB $p < 0.05$), ranked by band R^2 . 95% intervals are SBB bootstrap intervals. The final two columns report the band R^2 of the *same* cell when the target is switched to the quoted spread and to the Amihud price-impact proxy.

Firm	Cell	Band R^2	95% CI	SBB p	Verdict	Spread R^2
OXY	L3/core	0.157	[0.033, 0.235]	0.031	ROBUST	0.165
C	L3/core	0.118	[0.016, 0.261]	0.032	ROBUST	0.101
AMD	L2/act	0.083	[0.009, 0.152]	0.015	BORDERLINE	0.069
AAPL	L1/act	0.040	[0.008, 0.067]	0.018	ROBUST	0.033
USB	L1/core	0.039	[0.006, 0.074]	0.033	ROBUST	0.026
MSFT	L1/act	0.021	[0.002, 0.045]	0.046	ROBUST	0.019
CMCSA	L3/full	0.012	[0.002, 0.024]	0.040	ROBUST	0.011
META	L2/actC	0.010	[−0.001, 0.024]	0.047	BORDERLINE	0.010
DIS	L3/full	0.009	[−0.000, 0.020]	0.043	BORDERLINE	0.011

6.2 Horizon structure and liquidity channel

6.2.1 Lead time

Figure 3 traces the OOS R^2 of the headline cells against the forecast horizon h . The profiles share one shape: there is little or no accuracy — often slightly negative — at short horizons ($h \leq 6$ weeks), and what signal exists gathers in the pre-registered band $h \in [14, 21]$. A lead of roughly a quarter, rather than a few days, points to a slow channel: narrative geometry moves first, and residual illiquidity follows months later, which is hard to reconcile with a high-frequency mechanical story.

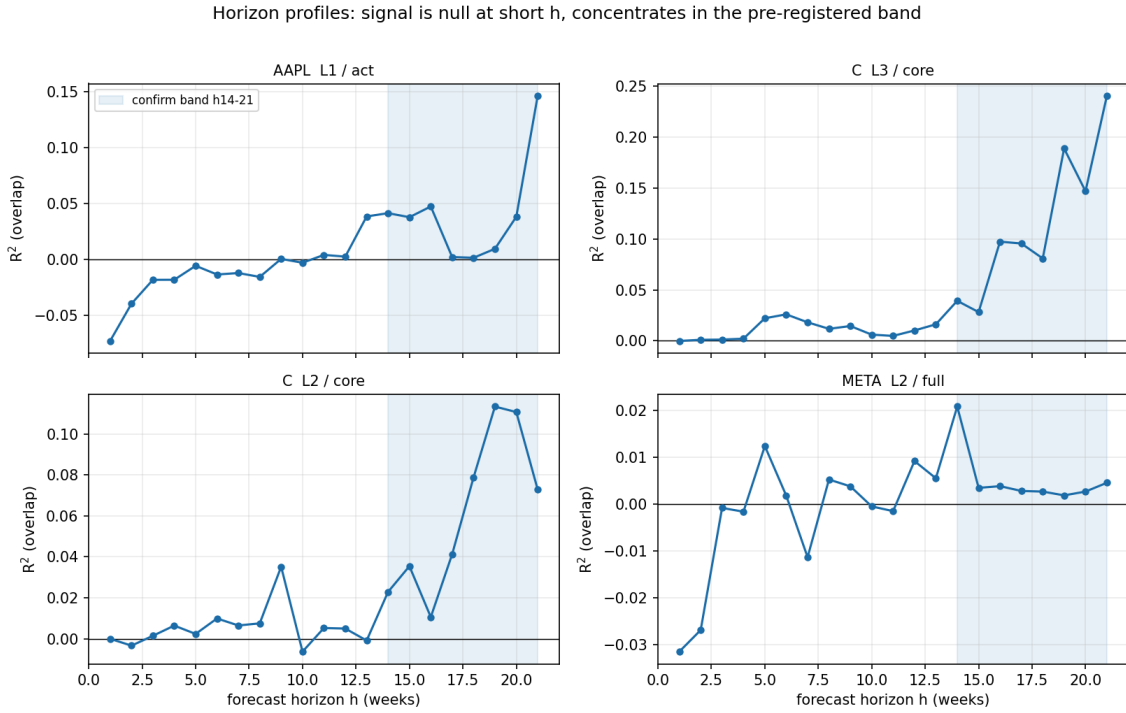


Figure 3: Horizon profiles for representative significant cells. The shaded region marks the confirmation band $h \in [14, 21]$. Accuracy is null at short horizons and concentrates in the band.

6.2.2 Which liquidity channel

This is the test of H_2 . To pin down the channel, we re-estimate each significant cell against all three illiquidity proxies: spread-per-volume (SV), the quoted bid–ask spread, and the Amihud price-impact ratio. Figure 4 shows the SV and quoted-spread band R^2 moving together and staying positive across the significant cells, while the Amihud R^2 hovers near zero or below; the final columns of Table 2 tell the same story, with the spread R^2 shadowing the SV R^2 . The signal thus loads on the quoting/adverse-selection part of illiquidity, not on the volume-scaled price-impact part — exactly the discriminating pattern SV was built to expose. Because SV divides the spread by dollar volume, a signal that survives in SV cannot be an artefact of the volume surge that accompanies disagreement trading; it resides in the adverse-selection widening of the spread itself, confirming H_2 .

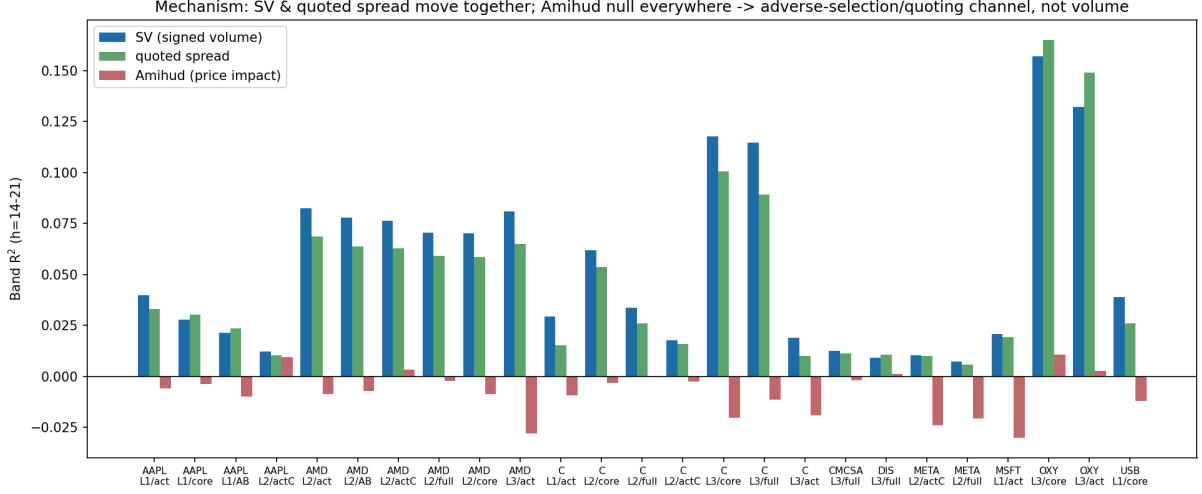


Figure 4: Band R^2 of each significant cell under three illiquidity proxies. Spread-per-volume (SV) and the quoted spread co-move and are positive; the Amihud price-impact proxy is null across cells, locating the effect in the quoting/adverse-selection channel.

6.3 Narrative episodes

6.3.1 Predictive episodes

When the model predicts well, it does so in short bursts rather than steadily. For each significant cell, we attribute the OOS R^2 to individual dates and group those dates into episodes; Table 3 reports the share of a cell’s total accuracy carried by its single largest episode and by its three largest. Across cells, the top three episodes account for between 0.66 and 1.00 of all the OOS accuracy. Each episode lines up, at the cell’s forecast lead h , with a datable news catalyst.

Table 3: Episodic decomposition of OOS accuracy. “Cell R^2 ” is the single-horizon localization estimate (computed at the listed lead h , min-train 156) and therefore differs from the band statistic of Table 2, which averages $h \in [14, 21]$. “Top-1” and “Top-3” give the fraction of the cell’s total attributed accuracy in its largest one and three episodes; “Catalyst” names the leading aligned event.

Firm	Cell R^2	h	Episodes	Top-1	Top-3	Leading catalyst
OXY	0.276	16	2	0.83	1.00	Icahn activism / Anadarko
AAPL	0.146	21	2	0.80	0.98	Apple–Epic antitrust ruling
CVX	0.091	20	2	0.62	0.99	Low-carbon strategy / COVID
AMD	0.082	18	5	0.37	0.66	Market-share / Xilinx deal
C	0.081	18	2	0.97	0.99	Restructuring / earnings

Figure 5 makes the lead/lag structure concrete for AMD. The forecast issued at t (top) is plotted against realized residual illiquidity (bottom), shifted forward by the cell’s lead of $h = 18$ weeks. The shaded bands are AMD’s predictive episodes from Table 3, and each numbered

marker ties an episode to a dated catalyst — the Xilinx acquisition and market-share gains of 2020–2021, the 2022 PC-demand downturn, and chip-export restrictions, and the 2023–2025 AI-accelerator cycle. Each episode aligns with a datable event rather than a quiet stretch. The gray region is the warm-up interval where no OOS forecast is defined.

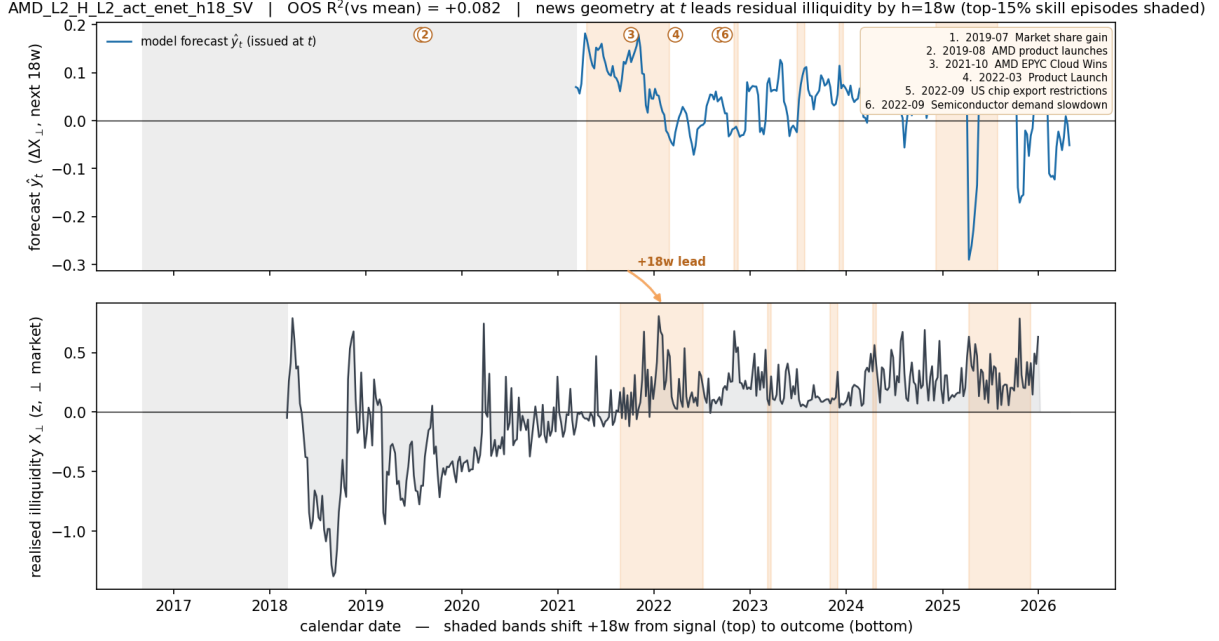


Figure 5: Lead/lag structure for AMD (L2/act, $h = 18$). Top: model forecast $\hat{y}_t$ of the change in residual illiquidity issued at t . Bottom: realized residual illiquidity $X_{\perp}$, with the predictive-episode bands shifted forward by $h = 18$ weeks. Numbered markers index the catalysts in the key.

6.3.2 The narrative geometry behind the episodes

The predictive episodes coincide with visible dislocations in the geometry. Figure 1 shows this for the inverse directional concentration $K_{t,k}^{-1}$: with each firm’s predictive episodes shaded onto its per-anchor trajectories, the shaded windows fall on marked movements in the feature paths rather than on quiet stretches. The predictable windows therefore track genuine reconfigurations of narrative structure, not estimation noise.

7 Discussion

H_1 holds conditionally, not pervasively. In aggregate the feature set does not beat the expanding-mean benchmark: the median band R^2 is indistinguishable from zero and about half of all cells are negative. The signal lives in the right tail—16 cells across nine firms that survive family-wise control—and within those cells it concentrates in a few datable episodes. Geometry predicts, but selectively: the measure is a detector of liquidity dislocations around large narrative events, not a continuous illiquidity forecaster.

H_2 locates that predictable variation in the adverse-selection component of the spread. The

signal predicts the quoted spread about as well as spread-per-volume (SV), while the Amihud price-impact proxy is null. SV and the raw spread move together; the volume-scaled Amihud ratio does not. This is the signature SV was built to expose: the predictable variation sits in the adverse-selection widening of the spread, not in the high-volume churn that ordinary disagreement trading also generates. It fits theories in which the arrival and reconfiguration of firm-level narratives change the information asymmetry faced by liquidity providers, who respond through quotes rather than through realized price impact. The horizon structure reinforces this reading: accuracy is null at short horizons and peaks at $h \in [14, 21]$ weeks, a slow channel at roughly a one-quarter lead that fits gradual diffusion of narrative information into microstructure and argues against a high-frequency mechanical explanation.

No single granularity dominates. The largest effects appear at the medium and coarse layers ($L2$, $L3$; e.g. OXY and Citigroup at $L3/\text{core}$), several smaller robust effects at the fine layer ($L1$; e.g. Apple, USB, and Microsoft). The appropriate narrative resolution is firm-dependent, which is why we retain the full layer hierarchy rather than fix one clustering granularity.

Three limitations qualify these conclusions. The robust set is small, so the cross-sectional claims rest on few firms. The forecast leads are long enough to admit confounding with slowly evolving fundamentals, and the residualization on the market-control block cannot fully exclude omitted common factors. And the alignment between episodes and named catalysts is interpretive: the per-date attribution establishes *when* the model predicts well, but the design does not identify a causal channel from a specific news event to the liquidity move. Establishing that causality, and extending the panel to sharpen the tail estimates, are the natural next steps.

A Memory-Window Calibration

This appendix documents how the per-layer memory windows — how far back each frame looks — were selected and verifies that the result does not hinge on that choice. Three checks agree: each layer’s narrative state has an intrinsic persistence that lengthens as the layer coarsens; a pre-registered window sweep finds the optimal half-life widening in step, $\approx 35 \rightarrow 53 \rightarrow 77$ sessions for the event, narrative, and macro layers, replicating on a second liquidity proxy; and the production pipeline therefore scores each layer on its own matched window, with a single global window kept only as a robustness comparison. The exploration proceeded in three phases.

Phase A (empirical persistence). Three independent estimators of narrative half-life were derived: the AR(1) and ACF 50%-crossing of the per-cluster activation a_{tk} ; the ACF 50%-crossing of the narrative-direction signal ϕ ; and the median run length of cluster-survival episodes (after thresholding soft activations at the per-cluster 75th percentile to avoid the degenerate “always-alive” artifact). Averaged across clusters, the three estimators converged on $\hat{\tau} \approx 7$ weeks (≈ 35 sessions), matching the event-layer window ($h = 35$, $\lambda \approx 0.02$); broken out by layer, persistence rises with coarseness, as the sweep below confirms.

Phase B (window sweep). Four log-spaced half-life candidates were evaluated:

$\{21, 35, 53, 77\}$ sessions (labeled short, incumbent, long, and longer). For each window, frames were rebuilt, and the full feature-construction and scoring pipeline was re-run, evaluated on out-of-sample elastic-net band- R^2 over the forward horizons of $h = 14$ –21 weeks. A best-of-family selector (activation-only vs. full-feature models) was applied per layer, and statistical comparisons used a Model Confidence Set (MCS, Hansen–Lunde–Nason 2011) together with a one-sided SPA test versus the incumbent. To guard against multiple-comparison inflation, the pre-registered metric and a capped candidate set were fixed before the sweep.

Phase C (selection and result). The sweep revealed that the optimal window scales with narrative granularity:

Table 4: Window sweep results by clustering layer (AAPL, SV proxy, $T = 322$ periods, $B = 2000$ bootstrap replications).

Layer	Optimal window	Half-life (sessions)	Best model	Band- R^2	SPA p
L1 (fine, $K \approx 16$ –40)	incumbent	35	act	0.040	0.012
L2 (medium, $K \approx 8$ –15)	long	53	act	0.083	0.013
L3 (coarse, $K \approx 3$ –7)	longer	77	full	0.022	0.047

The monotone rightward shift—optimal half-life increasing from ≈ 35 to ≈ 53 to ≈ 77 sessions as narratives become coarser—replicates exactly on a second liquidity proxy (bid-ask spread), confirming that the result is not a single-proxy artifact. The incumbent seven-week window lies in the MCS at all three layers and is optimal for L1 (the event level), while L2 and L3 are better served by the longer 53- and 77-session windows; the per-layer improvement is individually marginal in SPA terms ($p \approx 0.12$ –0.14), but it is monotone in layer coarseness and replicates on a second liquidity proxy, aligning each layer with its independently estimated

persistence (Phase A). Coupling the look-back to $\Delta_\ell = \lceil 4h_\ell \rceil$ (truncated tail mass $< 1/16$) removes the truncation confound that would otherwise contaminate the longer windows. The production pipeline therefore adopts the *per-layer matched windows*—half-lives 35, 53, and 77 sessions for L1, L2, and L3—scoring each layer on its own frames and retaining the single global window only as a robustness comparison.

B Financial-Discourse Projection: Details

This appendix details the projection summarized in Section 4.2.

Discourse direction removal. Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ denote the matrix of unit-normalized Gemini-001 embeddings for a ticker’s de-duplicated headlines, ordered by their original publication sequence (this ordering is preserved throughout the pipeline, as downstream modules index into it directly). A single randomized SVD pass on $\mathbf{X}$ extracts the dominant right singular vector

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{u}\|=1} \|\mathbf{X} \mathbf{u}\|^2. \quad (32)$$

This direction absorbs approximately 73% of the total Frobenius variance yet encodes only firm identity rather than thematic content [Mu and Viswanath, 2018]. Every embedding is then projected onto its orthogonal complement,

$$\mathbf{x}_i^\perp = \mathbf{x}_i - (\mathbf{x}_i \cdot \mathbf{v}_1) \mathbf{v}_1, \quad (33)$$

which collapses mean cosine similarity from ≈ 0.60 to < 0.10 , exposing the thematic structure that firm-identity anisotropy was masking.

Data-driven dimensionality selection. After discourse removal, the residual matrix $\mathbf{X}^\perp$ still inhabits a 768-dimensional ambient space, the majority of that variance being noise. Rather than fixing the number of retained components *a priori*, the pipeline selects the smallest k^* such that the cumulative spectral energy of $\mathbf{X}^\perp$ first crosses a desired target threshold τ :

$$k^* = \min \left\{ k : \frac{\sum_{j=1}^k \sigma_j^2}{\|\mathbf{X}^\perp\|_F^2} \geq \tau \right\}, \quad (34)$$

where $\sigma_1 \geq \sigma_2 \geq \dots$ are the singular values of $\mathbf{X}^\perp$. The denominator is the *true* Frobenius norm of the post-discourse-removal matrix rather than the sum of the retained singular values: normalizing by the truncated energy would force the cumulative variance to unity by construction and render the threshold inoperative. In practice, the elbow of the cumulative-variance curve is stable across the full corpus, and approximately 80% of the true Frobenius variance is recovered at convergence. The result is cached between runs, recomputed only when the

embedding model changes or an explicit full refit is requested. The randomized SVD of Halko et al. [2011] is used throughout for computational tractability on corpora of tens of thousands of headlines.

Projection and frozen-geometry mode. A second randomized SVD on $\mathbf{X}^\perp$ with k^* components yields the right singular matrix $\mathbf{V}_k \in \mathbb{R}^{D \times k^*}$ used in Eq. (7). In a frozen-geometry robustness mode designed for out-of-sample auditing, the basis $\mathbf{V}_k$ is fitted exclusively on headlines predating a fixed cut-off date; subsequent headlines are projected with the frozen basis and assigned to clusters without contributing to any fitting step, eliminating look-ahead leakage in the robustness analysis.

C Three-Layer Hierarchical Clustering: Details

This appendix details the clustering construction summarized in Section 4.2.3.

Shared graph geometry. Before any partition is computed, a shared affinity graph over $\bar{\mathbf{Y}}$ is constructed, and its spectral structure is computed once, then consumed by all three layers in four steps.

Near-duplicate collapse. Cross-day template headlines — e.g., recurring wire formats repeated verbatim across dates — form near-identical embedding families that within-day deduplication never eliminates. In the self-tuning affinity, such families acquire local scales near zero and exclusively internal nearest-neighbor edges, producing graph-disconnected components whose indicator functions dominate the bottom Laplacian eigenvectors. Therefore, the eigengap reports the number of template families rather than narrative themes. To prevent this, near-duplicate embeddings (cosine distance below a small threshold) are collapsed to a single graph node via union-find over their k -nearest-neighbor edges before the affinity is built. A mapping from original rows to graph nodes is retained so that partition labels can be broadcast back to the full corpus.

Self-tuning affinity. The affinity matrix follows Zelnik-Manor and Perona [2004]: each entry is

$$A_{ij} = \exp\left(-\frac{d_{ij}^2}{\sigma_i \sigma_j}\right), \quad (35)$$

where σ_i is the cosine distance from point i to its κ -th nearest neighbour and d_{ij} is the cosine distance between points i and j . The neighbourhood size κ scales with corpus size as $\kappa = \lfloor 7\sqrt{N/N_{\text{ref}}} \rfloor$, where N_{ref} is a reference corpus size calibrated once and frozen. This $\sqrt{N}$ scaling ensures that the effective neighbourhood spans the same conceptual radius at any corpus size, preventing the number of inferred clusters from inflating artificially as the corpus grows. A relative floor $\sigma_i \geq \delta \cdot \text{operatornamemedian}_j(\sigma_j)$ (with $\delta = 0.05$) prevents near-duplicate graph nodes from receiving $\sigma \approx 0$, which would zero all outward edge weights and disconnect those nodes from the rest of the graph. The resulting sparse matrix is turned symmetric by taking the element-wise maximum of A and $A^\top$.

Sinkhorn normalization. Iterative row- and column-scaling [Sinkhorn and Knopp, 1967] drives the sparse affinity to a doubly-stochastic matrix $\mathbf{C}$, equalizing the influence of every point on the graph regardless of local density. This addresses a known failure mode of density-based methods in high-dimensional spaces: in $\mathbb{R}^{201}$, covering density and cluster structure are nearly orthogonal, so density-based methods frequently conflate them.

Spectral decomposition. The bottom 64 eigenpairs of the normalized graph Laplacian

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{C} \mathbf{D}^{-1/2}, \quad (36)$$

where $\mathbf{D} = \text{diag}(\mathbf{C}\mathbf{1})$, are computed once via sparse Lanczos iteration. The eigenvalues are ascending; small eigenvalues correspond to slowly-varying Fiedler-like directions that encode cluster structure, while large eigenvalues correspond to high-frequency within-cluster variation. These eigenpairs are shared across all three layers and never recomputed.

Nested divisive partition. The production partitioner is *divisive* rather than agglomerative. Agglomerative linkage in high dimensions is susceptible to chaining artifacts in which a sequence of bridge points merges geometrically distant clusters; divisive spectral splitting is immune to this because each split is driven by the eigengap of the local affinity geometry rather than by inter-point distances evaluated across the full corpus.

L3 top-level partition. The number of macro-thematic clusters K_3 is selected by the eigengap heuristic [von Luxburg, 2007]: K_3 is the position of the most prominent gap in the ascending Laplacian spectrum within the window $[4, 10]$. Prominence is assessed on a scale-free basis: a gap is significant only if it is at least 1.5 times the median gap in the same window, so the decision carries no absolute threshold and is invariant to the overall scale of the spectrum. Spherical k -means [Dhillon and Modha, 2001] on the first K_3 eigenvectors (with rows normalized to unit length before clustering and centroid renormalization after each update) produces the L3 label vector $\ell^{(3)}$.

L2 divisive refinement. Each L3 parent cluster p is refined independently. Its share of the total corpus is $s_p = n_p/N$, and its permitted child-count band is $[k_{\text{lo}}, k_{\text{hi}}] = [\max(2, \lfloor K_{\text{lo}}^{(2)} s_p \rfloor), \min(8, \lceil K_{\text{hi}}^{(2)} \rceil)]$ where $[K_{\text{lo}}^{(2)}, K_{\text{hi}}^{(2)}] = [9, 23]$ is the layer’s global bound. For each parent, a fresh self-tuning affinity is constructed on the parent’s subset of $\bar{\mathbf{Y}}$ with corpus-size-adapted neighborhood parameters, Sinkhorn-normalized, and its within-parent eigengap selects the child count $k \in [k_{\text{lo}}, k_{\text{hi}}]$. Parents whose size falls below a minimum floor proportional to N are not split. The resulting L2 labels are a genuine sub-partition of $\ell^{(3)}$ by construction.

L1 divisive refinement. The identical procedure is applied to the L2 partition, targeting $K_1 \in [17, 56]$ with per-cluster floors scaled to the corpus size. Each L2 parent’s child-count band is again proportional to its share of N , with a maximum of eight children per parent as a safety rail to prevent any single parent from monopolizing the total cluster count.

Anchor-basis rank selection. For cluster k with member embeddings $\mathbf{Y}_k$, a truncated SVD on the mean-centered matrix $\mathbf{Y}_k - \bar{\mathbf{y}}_k$ retains the smallest number of right singular vectors

$\mathbf{B}_k \in \mathbb{R}^{k^* \times r_k}$ sufficient to explain at least a layer-specific variance threshold (90%, 85%, and 80% for L3, L2, and L1 respectively) of the *total* within-cluster Frobenius variance:

$$r_k = \min \left\{ r : \frac{\sum_{j=1}^r s_{k,j}^2}{\|\mathbf{Y}_k - \bar{\mathbf{y}}_k\|_F^2} \geq \tau_\ell \right\}, \quad (37)$$

where $s_{k,1} \geq s_{k,2} \geq \dots$ are the within-cluster singular values and τ_ℓ is the layer-specific threshold. The denominator is the true total within-cluster variance, not the sum of the retained singular values; the same honesty principle applies here as in Eq. (34). Per-layer rank ceilings (60, 25, and 15 for L3, L2, and L1) reflect the intuition that macro-thematic clusters have richer internal geometry, while event-level clusters are tightly focused.

Temperature calibration. The concentration parameter α of Eq. (9) is calibrated per layer by sweeping a candidate grid $\alpha \in \{0.5, 1, 1.5, 2, 3, 4, 6, 8, 10, 12, 15, 18, 22, 28, 35, 45, 60\}$ and selecting the smallest value that satisfies two conditions jointly: (i) the soft weight entropy $H = -\sum_k \bar{w}_k \log \bar{w}_k$ falls within a layer-specific effective-anchor-count target range (equivalent to $K_{\text{eff}} = e^H \in [1.4, 2.2], [2.0, 4.5], [2.5, 6.0]$ for L3, L2, L1 respectively), and (ii) at least 80%, 80%, and 75% of headlines, for L3, L2, and L1 respectively, have a dominant soft weight above a minimum threshold. This joint criterion prevents mode collapse — all weight on a single cluster — and over-diffusion — weights too flat to discriminate. If no candidate satisfies both criteria, the value closest to the midpoint of the entropy target range is selected as a fallback.

D Headline Preprocess Prompt

Filter and clean news about entity for semantic analysis. HEADLINE: "title" SNIPPET: "snippet" TASK: Extract ALL distinct, semantically rich business facts from the combined headline and snippet that are temporally aligned with the publication timestamp. Each fact must be self-contained, mutually distinct, and suitable for embedding-based clustering. Do not extract multiple phrasings of the same underlying event—only genuinely separate facts or causal chains. Do not include facts that merely reference past events without new information. KEEP — concrete business drivers with causal/relational depth: - Operational metrics (revenue, margins, unit volumes) with magnitude context - Strategic actions (acquisitions, partnerships, restructuring) with rationale - Supply chain causality (tariff $\rightarrow$ cost increase $\rightarrow$ pricing response) - Temporal anchors (Q1 2025, fiscal deadlines, launch timelines) - Competitive positioning (market share shifts, entry/exit decisions) - Regulatory dependencies (antitrust rulings, patent decisions with business impact) - Executive changes with strategic context DISCARD — low-information noise that collapses embedding clusters: - Stock prices, price targets, ratings ("PT raised to \$275", "shares up 3%") - Market cap milestones, trading volume commentary - Generic advice ("should you buy", "top picks") - Vague clickbait without operational specifics ("Here's why...") -

Listicle mentions without entity-specific substance EXTRACTION RULES: - Emit one entry per distinct event, causal chain, or operational driver present in the text - Each fact must be self-contained (requires no cross-referencing with other facts) - Preserve causal chains (X enables Y) and temporal structure - Include specific numbers, dates, actor names - Strip stock-price framing ("shares surge after..." : keep the "after") - Max 250 characters per fact - No invented details OUTPUT: JSON "discarded": boolean, "facts": ["fact 1", "fact 2", ...]

References

- Daron Acemoglu, Victor Chernozhukov, and Muhamet Yildiz. Learning and disagreement in an uncertain world. NBER Working Paper 12648, National Bureau of Economic Research, 2006.
- Yakov Amihud. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002. doi: 10.1016/S1386-4181(01)00024-6.
- Yakov Amihud and Haim Mendelson. Asset pricing and the bid-ask spread. *Journal of Financial Economics*, 17(2):223–249, 1986. doi: 10.1016/0304-405X(86)90065-6.
- Yakov Amihud, Andrew Karolyi, and Joonki Noh. Illiquidity and stock returns ii: Cross-section and time-series effects. *Review of Financial Studies*, 34(4):2101–2123, 2021. doi: 10.1093/rfs/hhaa080.
- Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003. doi: 10.1111/1468-0262.00418.
- Dogu Araci. Finbert: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*, 2019.
- Adem Atmaz and Suleyman Basak. Belief dispersion in the stock market. *Journal of Finance*, 73(3):1225–1279, 2018. doi: 10.1111/jofi.12618.
- Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636, 2016. doi: 10.1093/qje/qjw024.
- Arindam Banerjee, Inderjit S. Dhillon, Joydeep Ghosh, and Suvrit Sra. Clustering on the unit hypersphere using von mises-fisher distributions. *Journal of Machine Learning Research*, 6: 1345–1382, 2005.
- Snehal Banerjee. Learning from prices and the dispersion in beliefs. *Review of Financial Studies*, 24(9):3025–3068, 2011. doi: 10.1093/rfs/hhr050.
- Snehal Banerjee and Ilan Kremer. Disagreement and learning: Dynamic patterns of trade. *Journal of Finance*, 65(4):1269–1302, 2010. doi: 10.1111/j.1540-6261.2010.01570.x.
- Tim Bollerslev, Jia Li, and Yuan Xue. Volume, volatility, and public news announcements. *Review of Economic Studies*, 85(4):2005–2041, 2018. doi: 10.1093/restud/rdy003.
- Leland Bybee, Bryan T. Kelly, and Yinan Su. Narrative asset pricing: Interpretable systematic risk factors from news text. *Review of Financial Studies*, 36(12):4759–4787, 2023. doi: 10.1093/rfs/hhad042.
- Leland Bybee, Bryan T. Kelly, Asaf Manela, and Dacheng Xiu. Business news and business cycles. *Journal of Finance*, 79(5):3105–3147, 2024. doi: 10.1111/jofi.13377.

- H. Henry Cao and Hui Ou-Yang. Differences of opinion of public information and speculative trading in stocks and options. *Review of Financial Studies*, 22(1):299–335, 2009. doi: 10.1093/rfs/hhn046.
- Bruce I. Carlin, Francis A. Longstaff, and Kyle Matoba. Disagreement and asset prices. *Journal of Financial Economics*, 114(2):226–238, 2014. doi: 10.1016/j.jfineco.2014.06.007.
- Yifei Chen, Bryan T. Kelly, and Dacheng Xiu. Expected returns and large language models. *Review of Financial Studies*, 2025. SSRN Working Paper 4416687.
- Julio A. Crego. Why does public news augment information asymmetries? *Journal of Financial Economics*, 137(1):72–89, 2020. doi: 10.1016/j.jfineco.2020.02.005.
- Gilles Deleuze and Félix Guattari. *Anti-Oedipus: Capitalism and Schizophrenia*. University of Minnesota Press, Minneapolis, 1983. Trans. R. Hurley, M. Seem, and H. R. Lane.
- Norman K. Denzin. *Interpretive Biography*, volume 17 of *Qualitative Research Methods*. Sage Publications, Newbury Park, CA, 1989.
- Inderjit S. Dhillon and Dharmendra S. Modha. Concept decompositions for large sparse text data using clustering. *Machine Learning*, 42(1):143–175, 2001. doi: 10.1023/A:1007612920971.
- Karl B. Diether, Christopher J. Malloy, and Anna Scherbina. Differences of opinion and the cross section of stock returns. *Journal of Finance*, 57(5):2113–2141, 2002. doi: 10.1111/0022-1082.00490.
- Zihan Dong, Xinyu Fan, and Zhiyuan Peng. Fnspid: A comprehensive financial news dataset in time series, 2024.
- David Easley, Nicholas M. Kiefer, Maureen O’Hara, and Joseph B. Paperman. Liquidity, information, and infrequently traded stocks. *Journal of Finance*, 51(4):1405–1436, 1996. doi: 10.1111/j.1540-6261.1996.tb04074.x.
- David Easley, Søren Hvidkjaer, and Maureen O’Hara. Is information risk a determinant of asset returns? *Journal of Finance*, 57(5):2185–2221, 2002. doi: 10.1111/1540-6261.00493.
- David Easley, Marcos M. López de Prado, and Maureen O’Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5):1457–1493, 2012. doi: 10.1093/rfs/hhs053.
- Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2):303–353, 1998. doi: 10.1137/S0895479895290954.
- Larry G. Epstein and Martin Schneider. Ambiguity, information quality, and asset pricing. *Journal of Finance*, 63(1):197–228, 2008. doi: 10.1111/j.1540-6261.2008.01314.x.

- Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65, 2019. doi: 10.18653/v1/D19-1006.
- Financial Modeling Prep. Financial modeling prep api documentation. <https://site.financialmodelingprep.com/developer/docs/>. Accessed 2026-06-17.
- Xavier Gabaix, Ralph S. J. Koijen, Robert J. Richmond, and Motohiro Yogo. Asset embeddings. NBER Working Paper 33651, National Bureau of Economic Research, 2025.
- Lawrence R. Glosten and Lawrence E. Harris. Estimating the components of the bid/ask spread. *Journal of Financial Economics*, 21(1):123–142, 1988. doi: 10.1016/0304-405X(88)90034-7.
- Lawrence R. Glosten and Paul R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100, 1985. doi: 10.1016/0304-405X(85)90044-3.
- Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011. doi: 10.1137/090771806.
- William L. Hamilton, Jure Leskovec, and Dan Jurafsky. Diachronic word embeddings reveal statistical laws of semantic change. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1489–1501, 2016. doi: 10.18653/v1/P16-1141.
- Milton Harris and Artur Raviv. Differences of opinion make a horse race. *Review of Financial Studies*, 6(3):473–506, 1993. doi: 10.1093/rfs/6.3.473.
- Joel Hasbrouck. Trading costs and returns for u.s. equities: Estimating effective costs from daily data. *Journal of Finance*, 64(3):1445–1477, 2009. doi: 10.1111/j.1540-6261.2009.01469.x.
- Harrison Hong and Jeremy C. Stein. Disagreement and the stock market. *Journal of Economic Perspectives*, 21(2):109–128, 2007. doi: 10.1257/jep.21.2.109.
- Chong Huang, Radhika Lunawat, and Qiguang Wang. Disagreement about public information quality and informational price efficiency. *Journal of Financial Economics*, 152:103762, 2024. doi: 10.1016/j.jfineco.2023.103762.
- Roger D. Huang and Hans R. Stoll. The components of the bid-ask spread: A general approach. *Review of Financial Studies*, 10(4):995–1034, 1997. doi: 10.1093/rfs/10.4.995.
- Eugene Kandel and Neil D. Pearson. Differential interpretation of public signals and trade in speculative markets. *Journal of Political Economy*, 103(4):831–872, 1995. doi: 10.1086/262005.
- Zheng Tracy Ke, Bryan T. Kelly, and Dacheng Xiu. Predicting returns with text data. NBER Working Paper 26186, National Bureau of Economic Research, 2019.

- Oliver Kim and Robert E. Verrecchia. Market liquidity and volume around earnings announcements. *Journal of Accounting and Economics*, 17(1–2):41–67, 1994. doi: 10.1016/0165-4101(94)90004-3.
- Oliver Kim and Robert E. Verrecchia. Pre-announcement and event-period private information. *Journal of Accounting and Economics*, 24(3):395–419, 1997. doi: 10.1016/S0165-4101(98)00013-5.
- Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985. doi: 10.2307/1913210.
- Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999. doi: 10.1103/PhysRevLett.83.1467.
- Sophia Zhengzi Li and Zeyao Luan. News-based investor disagreement and stock returns. *Review of Accounting Studies*, 30:2312–2375, 2025. doi: 10.1007/s11142-025-09897-1.
- Zhuang Liu, Degen Huang, Kaiyu Huang, Zhuang Li, and Jun Zhao. Finbert: A pre-trained financial language representation model for financial text mining. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4513–4519, 2020. doi: 10.24963/ijcai.2020/622.
- Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *Journal of Finance*, 66(1):35–65, 2011. doi: 10.1111/j.1540-6261.2010.01625.x.
- Tim Loughran and Bill McDonald. Textual analysis in accounting and finance: A survey. *Journal of Accounting Research*, 54(4):1187–1230, 2016. doi: 10.1111/1475-679X.12123.
- Asaf Manela and Alan Moreira. News implied volatility and disaster concerns. *Journal of Financial Economics*, 123(1):137–162, 2017. doi: 10.1016/j.jfineco.2016.01.032.
- Deborah Miori and Constantin Petrov. Narratives from GPT-derived networks of news and a link to financial markets dislocations. *International Journal of Data Science and Analytics*, 2024. doi: 10.1007/s41060-024-00516-x.
- Stephen Morris. The common prior assumption in economic theory. *Economics and Philosophy*, 11(2):227–253, 1995. doi: 10.1017/S0266267100003382.
- Jiaqi Mu and Pramod Viswanath. All-but-the-top: Simple and effective postprocessing for word representations. In *International Conference on Learning Representations (ICLR)*, 2018.
- Ľuboš Pástor and Robert F. Stambaugh. Liquidity risk and expected stock returns. *Journal of Political Economy*, 111(3):642–685, 2003. doi: 10.1086/374184.
- Vasiliki Plerou, Parameswaran Gopikrishnan, Bernd Rosenow, Luís A. N. Amaral, and H. Eugene Stanley. Universal and non-universal properties of cross-correlations in financial time series. *Physical Review Letters*, 83(7):1471–1474, 1999. doi: 10.1103/PhysRevLett.83.1471.

- Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *Proceedings of the 15th European Signal Processing Conference (EUSIPCO)*, pages 606–610, 2007.
- Robert J. Shiller. Narrative economics. *American Economic Review*, 107(4):967–1004, 2017. doi: 10.1257/aer.107.4.967.
- Robert J. Shiller. *Narrative Economics: How Stories Go Viral and Drive Major Economic Events*. Princeton University Press, Princeton, NJ, 2019. ISBN 9780691182292.
- Richard Sinkhorn and Paul Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- Suvrit Sra. A short note on parameter approximation for von mises-fisher distributions: And a fast implementation of $i_s(x)$. *Computational Statistics*, 27(1):177–190, 2012. doi: 10.1007/s00180-011-0232-x.
- Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3):1139–1168, 2007. doi: 10.1111/j.1540-6261.2007.01232.x.
- Ulrike von Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416, 2007. doi: 10.1007/s11222-007-9033-z.
- Robert E. Whaley. The investor fear gauge. *Journal of Portfolio Management*, 26(3):12–17, 2000. doi: 10.3905/jpm.2000.319728.
- Yi Yang, Mark Christopher Siy Uy, and Allen Huang. FinBERT: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*, 2020.
- Lihi Zelnik-Manor and Pietro Perona. Self-tuning spectral clustering. In *Advances in Neural Information Processing Systems (NIPS)*, volume 17, pages 1601–1608, 2004.